

A Large-Scale Evaluation of Merger Simulations

VIVEK BHATTACHARYA^{1,2}, GASTÓN ILLANES^{1,2}, AVNER A. KREPS¹,
JOSÉ D. SALAS³, and DAVID STILLERMAN⁴

¹Department of Economics, Northwestern University

²National Bureau of Economic Research

³Cornerstone Research

⁴Kogod School of Business, American University

December 19, 2025

ABSTRACT. Prospective merger simulations are a commonly used tool in antitrust, but evidence about their accuracy is limited. We study 92 consummated mergers in consumer packaged goods and compare the realizations of price changes with predictions from merger simulations. Predicted price changes are correlated with, but typically larger than, realized price changes. Merger simulations have limited predictive power but are still typically more predictive than relying solely on structural presumptions. We use the departure between realization and predictions to estimate supply-side changes after the merger and find a role for both synergies and a departure from Bertrand-Nash conduct post-merger.

KEYWORDS. Antitrust, Merger Simulation.

JEL CODES. D43, K21, L13, L41.

CONTACT. vivek.bhattacharya@northwestern.edu, gaston.illanes@northwestern.edu, avner@u.northwestern.edu, jdsalas@u.northwestern.edu, stillerman@american.edu. We would like to acknowledge Igal Hendel, JF Houde, Alessandro Iaria, Sarah Johnston, Lorenzo Magnolfi, Rob Porter, and Chris Sullivan for useful feedback. We are grateful for excellent assistance from numerous research assistants; a list can be found in the acknowledgments in Bhattacharya et al. (2025). This project was funded by grants from the Center for Equitable Growth and the National Science Foundation (SES-2116934). Researcher(s) own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein. José D. Salas worked on this paper while a PhD student at Northwestern University. All errors are our own.

I. Introduction

Antitrust agencies frequently make decisions about whether to challenge proposed mergers and thus must make prospective predictions of their effects. A commonly used tool to guide these decisions is a merger simulation. Merger simulations have become a workhorse tool in industrial organization and are prominent in both the classroom and the courtroom (Werden and Froeb, 2006). However, despite repeated calls for it (Whinston, 2007; Angrist and Pischke, 2010; Nevo and Whinston, 2010), there is very little work comparing the predictions from these simulations to realized price changes of mergers.

Prior work that has tackled this question has typically focused on a small handful of mergers at a time. This paper does so in the context of almost 100 mergers in US consumer packaged goods (CPG) from 2006 to 2017. These mergers are a large subset of all transactions with a deal size larger than \$280 million involving CPG products sold through retail outlets. For each merger in our sample, we run every step of a textbook merger simulation (Hausman et al., 1994; Werden and Froeb, 1994; Nevo, 2000) using state-of-the-art methods. We first collect product characteristics and cost shifters for each market and estimate a mixed logit demand system (Berry et al., 1995) using pre-merger data on prices and quantities sold. We invert the first-order conditions from a Bertrand-Nash model to recover implied marginal costs, and we project these recovered costs onto cost shifters using nonparametric methods. We finally forecast demand and costs into future periods and solve the first-order conditions of the Bertrand-Nash game with the post-merger ownership structure. This procedure gives us predicted prices for about 1,000 markets (geography-month pairs) for each merger, typically with dozens of products per merger.

One main contribution of this scale is that we are able to identify systematic patterns in the comparisons between merger simulations and realizations of prices. First, we find that predicted price changes are directionally informative of actual price changes: a regression of actual price changes for merging parties on predicted price changes from the simulation, aggregated to the merger, returns a coefficient of 0.61 (s.e. 0.17). This slope is not just due to changes in marginal costs due to observed cost shifters: we find a significant relationship even when controlling for our prediction of changes in costs as well as measures of market structure. Moreover, larger predicted changes in merging party prices are correlated with larger realized changes at more granular levels as well—and this result holds even within-merger. Coefficients are typically smaller but still statistically significant when considering overall price changes or those of non-merging parties.

Second, a related finding is that predictions from merger simulations are systematically biased: merger simulations overpredict the price increase on about 75% of mergers. The average overprediction is about 2.3 pp. This overprediction increases over time and is thus not due to a delay in adjusting to the next equilibrium.

Third, we find that merger simulations are somewhat predictive of price changes: the R^2 of a regression of actual price changes on predicted price changes for merging parties, at the merger level, is 27%. Notably, this is larger than the predictive power of market structure, even controlling for cost shifters. Moreover, merger simulations have similar predictive power at the geography level and have significantly larger predictive power than market structure measures within-merger across geographies. Predictive power for non-merging party price changes is lower, though still comparable to the predictive power of market structure—and more predictive of price changes within-merger.

One way to interpret these results is through the lens of an analyst using simulations for prospective evaluation of mergers. These results provide some support for the use of merger simulations as an out-of-the-box tool in practice, although the at best moderate predictive power still highlights the role of other forms of analysis. A second lens is that of an economist interested in using the departure between predictions and realizations to learn about other effects of mergers. We find limited support for changes to demand driving the prediction error document above, and we thus use these results to understand the “supply-side residual.”

Under the assumption that Bertrand-Nash conduct holds after the merger, this prediction error is attributable to changes in marginal costs. Maintaining this assumption, we find that average synergies for merging parties amount on average to 2.7% of the initial price of the product; approximately 75% of mergers realize synergies on average. Moreover, some patterns in the data are at least heuristically consistent with these “implied synergies” being consistent with true changes in marginal cost. First, one would not expect non-merging parties to directly realize synergies: these implied synergies are indeed much smaller for non-merging parties and average less than 1%. Second, the extent of synergies increases over time since merger completion, consistent with operational changes (or contract renegotiation, if synergies are from increased buyer power) taking time to materialize. Third, consistent with distribution improvements leading to synergies, synergies for merging parties are correlated across products within-geography: DMAs explain 24% of the variation in synergies across merger-DMA-product groups. Synergies are also clustered across geographies within-product (with an R^2 of 0.28), consistent with operational changes at the product level.

We also find, however, that under the Bertrand-Nash assumption, the implied synergy is systematically larger when the unilateral effect of the merger—the incentive for all parties to change price due to the merger—is larger. There are some plausible explanations for this correlation at certain levels of analysis: for instance, if agencies expect an especially large incentive to increase price post-merger, they may expect the merger to generate synergies in order to avoid a challenge. We discuss potential reasons for the correlation at other levels, but we find them a priori unlikely within-merger and view this correlation as an indication that the Bertrand-Nash assumption may

be inappropriate. We thus use this pattern to construct a conduct test based on the assumption that the post-merger conduct is such that the implied synergies (controlling for product observables and cost shifters) are uncorrelated with changes in markups from the model. We use this moment condition to estimate a imperfect profit internalization parameter between merging parties. While there is significant variation in this conduct parameter across mergers, we find frequent departures from Bertrand-Nash conduct post-merger. On average, merging parties only internalize slightly less than half of the profits of their new partners. Putting all these results together, we conclude that about half of the prediction error in merger simulations are attributable to synergies and half to (this version of) departures from the standard conduct assumptions.

Related Literature. The literature comparing predictions from merger simulations to realizations is sparse, and each paper in this literature studies a very small handful of mergers. Nevo (2000) studies two ready-to-eat cereal mergers, Peters (2006) studies five airline mergers, Weinberg (2011) and Weinberg and Hosken (2013) together study three mergers of consumer goods, Björnerstedt and Verboven (2016) studies one merger of pharmaceuticals, and Miller and Weinberg (2017) study one joint venture in beer.¹ With the exception of Nevo (2000) and one merger in Weinberg and Hosken (2013), most simulations depart significantly from the realizations. Reasons for the departure are varied and include unexplained supply-side issues (Peters, 2006), potential issues in demand estimation (Weinberg and Hosken, 2013; Björnerstedt and Verboven, 2016), and post-merger coordination (Björnerstedt and Verboven, 2016; Miller and Weinberg, 2017). Relative to this literature, our main contribution is scale: by studying a large set of mergers—and both the cross-merger and within-merger heterogeneity in results—we can find systematic patterns.

The literature on synergies attributable to mergers is also sparse: Asker and Nocke (2021, p. 221) note that “[e]vidence as to the typical size of merger efficiencies is also extremely limited.” The extant literature can be organized into three groups. Merger retrospectives (Kim and Singal, 1993; Focarelli and Panetta, 2003; Bhattacharya et al., 2025) often attribute decreases in prices to synergies but do not provide a quantification of them. A second set of papers retrospectively estimates direct proxies for synergies, such as distribution distance in beer (Ashenfelter et al., 2015), heat rate in power plants (Demirer and Karaduman, 2025), inventory and capacity utilization in manufacturing (Braguinsky et al., 2015), or costs in hospitals (Schmitt, 2017). The final approach is closest to ours: some of the aforementioned papers evaluating merger simulations provide implied synergies

¹Garmon (2017) is distinct in both the focus on hospital mergers and in the quantity: he studies 28 mergers and evaluates a number of screening tools, although he assesses “reduced-form” merger simulations. Similar papers include Fournier and Gai (2007) and May and Noether (2014), which evaluate other reduced-form screens using in merger analysis, and Dranove and Ody (2016), which evaluates the effectiveness of these screens in predicting the cross-sectional variation in prices. Finally, though it does not evaluate merger simulation directly, Panhans and Taragin (2023) emphasizes the effects that a potentially misspecified demand model can have on the outcome of merger simulation and merger review.

required to rationalize the price effect as an auxiliary result.² We take a similar approach—although we interact it with our conduct estimation—and are able to provide estimates of synergies on a much broader scale than previous papers in the literature.

Conduct estimation has a long tradition in industrial organization (Porter, 1983; Bresnahan, 1987), and there has been renewed interest in it (Miller and Weinberg, 2017; Sullivan, 2020; Backus et al., 2021; Dearing et al., 2024; Duarte et al., 2024). We explore a different moment condition than typically used in the literature, taking advantage of the variation in our dataset. Moreover, relative to this literature, we aim to document patterns in conduct in a broader set of markets than is typical.

A recent set of papers (Brand, 2021; Döpper et al., 2025; Atalay et al., 2025) performs large-scale demand estimation using scanner data for consumer packaged goods but tackle a different question: they document evolution of markups over time. They also either devise or implement new methods for demand estimation: Brand (2021) uses the “FRAC” estimator from Salanié and Wolak (2022), Döpper et al. (2025) uses covariance restrictions from MacKay and Miller (2025) for identification rather than traditional instruments, and Atalay et al. (2025) use nested logit but propose a new data-driven method for constructing nests. While we find these approaches promising, we intentionally use more established methods for demand estimation and choices for instruments in an effort to evaluate merger simulations as they are likely to be done currently in practice. By collecting data on characteristics and plausible instruments, we also avoid some of the data issues that these newer methods are designed to address.

Moving beyond merger simulations, despite the rapidly growing importance of structural modeling in empirical economics, there are few tests of the accuracy of counterfactual predictions. A limited set of examples include an analysis of microfinance (Kaboski and Townsend, 2011), predictions of auctions models (Bajari and Hortacsu, 2005; Athey et al., 2011), discrete choice models (Pathak and Shi, 2021), and trade (Kehoe et al., 2017; Adão et al., 2025). We believe more work is warranted, and we view this paper as a contribution to this literature.

II. Data and Sample Selection

II.A. Data

Our main source of data for demand estimation is from NielsenIQ (2025). NielsenIQ provides a Retail Scanner Dataset, which provides prices and quantities at the UPC-store-week level for about 2.6–4.5 million UPCs, depending on the sample year. The dataset covers food, non-food grocery items, health and beauty aids, and select general merchandise. It also has fairly comprehensive

²Some papers cannot be classified into these bins. Grieco et al. (2018) estimates returns to scale from a production function, makes predictions of these efficiencies based on expected changes to quantities, and verifies these predictions with post-merger data in the context of beer. Jeziorski (2014) use the decision to merge as a way to estimate expected synergies.

coverage of retail outlets in the United States, although a handful of large retailers are known to be excluded from the sample.³ NielsenIQ provides an additional dataset called the Consumer Panel, which tracks shopping behavior by individual households over time and provides properties of these households. We have access to this sample from 2006 to 2018.

We collect external sources of data to supplement the NielsenIQ dataset on three dimensions relevant for demand estimation. First, since NielsenIQ does not provide ownership of each product, we obtain this information from a combination of Euromonitor Passport (Euromonitor International, 2022) and manual searches. Second, we collect potential cost shifters for each merger by listing inputs (e.g., wheat for cereal) and obtaining commodity price indices, typically from Federal Reserve Economic Data (FRED) from the Federal Reserve Bank of St. Louis (2025). Third, the NielsenIQ dataset has limited information about product characteristics, with variables that differ at the product module level. We supplement this dataset with additional product characteristics. Some of these characteristics are from unstructured searches, but many are from decoding the abbreviations and information embedded in the NielsenIQ UPC description. Table B.1 includes a list of the instruments and product characteristics used in each merger.

II.B. Merger Selection, Market Definition, and Product Definition

NielsenIQ’s data structure provides additional information that is useful for demand estimation. First, Nielsen defines a unit of geography called a designated market area (DMAs), which is a collection of counties that is subject to the same local television stations. We follow other papers that conduct demand estimation (Nevo, 2000; Miller and Weinberg, 2017) and use the DMA to construct a geographic market. Second, NielsenIQ assigns products to “modules.” We typically use modules as our definition of a product markets, a decision that is aligned with other papers that do demand estimation at scale (Atalay et al., 2025; Döpper et al., 2025). However, we have found through manual inspection that it is sometimes reasonable to combine modules together: for instance, we combine the “Wine-Domestic Dry Table” and “Wine-Imported Dry Table” modules. In a small handful of cases, we also found it reasonable to split modules based on characteristics when we think that products in each split are unlikely to be substitutes.⁴ Table B.1 provides the full list of product markets used in our analysis. As discussed in Bhattacharya et al. (2025), these market definitions seem consistent with the so-called *Brown Shoe* factors cited in the 2023 Merger Guidelines and are also consistent with definitions used historically by the DOJ and FTC. We also

³NielsenIQ describes this dataset as providing “scanner data from 35,000 to 50,000 grocery, drug, mass merchandise, and other stores, covering more than half the total sales volume of US grocery and drug stores and more than 30 percent of all US mass merchandiser sales volume.”

⁴For instance, we split “Baby Milk and Milk Flavoring” by whether the formula is for infants. The “Seasoning-Dry” module contains many different types of spices, such as garlic powder and cinnamon, which likely do not compete with each other. Dog and cat treats likely do not compete with each other in the “Dog and Cat Treats” module.

believe that our definitions of markets are likely to contain most relevant relevant substitutes for the products.

Our goal is to obtain a relatively large sample of mergers that are representative of typical mergers in consumer packaged goods. To do so, we follow a procedure very similar to the one in Bhattacharya et al. (2025). We start with a set of deals tracked by SDC Platinum from Thomson Reuters (2021), which provides comprehensive information on mergers, acquisitions, and joint ventures, and we restrict to deals that involve mergers of manufacturers of products sold in groceries and mass merchandisers. We consider every such deal with a transaction size larger than \$280 million, and we define a merger to be a combination of a deal and a product market. We only keep mergers where both the target and the acquirer sold at least one UPC in the same market with non-negligible market share—which ensures that we retain national, seasonal, and important regional products.⁵ This leads to 123 mergers over 44 deals. Finally, for computational reasons, we filter each merger to the top 50 DMAs by sales volume.

We must also take a stance on the definition of a product in our sample. Papers differ in whether the definition of a product is the UPC (Atalay et al., 2025), the brand (Döppler et al., 2025), or some level of aggregation between the two (Backus et al., 2021). The granularity of brands differs significantly across modules in the Nielsen data, so we also adopt an intermediate approach: in most markets our definition of a product is at the unique brand-characteristics level, but we sometimes define a product as a UPC in an effort to homogenize the granularity of products across product markets. For instance, the “Chili-Shelf Stable” module has 343 unique UPCs, only 215 of which have unique brand-characteristics, while the “Stew - Beef - Shelf Stable,” “Stew - Chicken - Shelf Stable,” and “Stew - Remaining - Shelf Stable” modules (which we group into one product market) have 98 UPCs, 86 of which have unique brand-characteristics, despite the ex-ante similarity of the characteristics we define on each.⁶ We therefore aggregate within brand-characteristics for chili but keep stew at the UPC level.

II.C. Descriptive Statistics

Panel A of Table 1 reports summary statistics for our initial sample. The markets in our sample are fairly concentrated, with the median Herfindahl-Hirschman Index (HHI) around 2,700.⁷ Many

⁵Appendix D of Bhattacharya et al. (2025) shows that this sample selection procedure leads of high coverage of UPCs within each product market.

⁶In addition to package size and store brand, which are used for every product market, we define “meat type,” “is hot,” “has beans,” and “chili con carne” characteristics for the chili product market, and “Brunswick,” “contains beef,” “contains chicken,” “microwave,” and “regular flavor” characteristics for the stew product market.

⁷The HHI is the sum of the squared market shares of each firm, and we compute this using the inside shares. We compute the change in the HHI (which we refer to as the DHHI) as the naive, or pro-forma, HHI, assuming that the market shares of merged entities would be the sum of the pre-merger market shares. Without divestitures, this is simply twice the product of the pre-merger market shares of the merging parties. We scale both these from 0 to 10,000.

			Quantiles				
	Mean	St. Dev.	10%	25%	50%	75%	90%
A. Initial Sample ($N = 123$)							
HHI	2,984	1,350	1,546	2,031	2,686	3,721	4,661
DHHI	158.6	332.5	0.3	2.0	18.1	154.6	418.1
$\mathbb{1}[\text{DHHI} > 100]$	0.33	0.47	0	0	0	1	1
Merging Party Share	0.20	0.17	0.03	0.06	0.15	0.30	0.48
Avg. Market Size (\$1,000s)	575	865	29	71	206	644	1,607
Number of Products	77	45	24	41	74	106	145
Share in Department							
Dry Grocery	0.60						
Health & Beauty Care	0.16						
Alcoholic Beverages	0.12						
B. Estimation Sample ($N = 92$)							
HHI	2,981	1,345	1,537	2,023	2,614	3,740	4,609
DHHI	137.9	284.2	0.1	1.5	14.9	135.9	339.7
$\mathbb{1}[\text{DHHI} > 100]$	0.29	0.46	0	0	0	1	1
Merging Party Share	0.19	0.18	0.02	0.05	0.13	0.27	0.49
Avg. Market Size (\$1,000s)	615	934	29	64	230	639	1,621
Number of Products	76	45	23	41	73	104	131
Share in Department							
Dry Grocery	0.62						
Health & Beauty Care	0.08						
Alcoholic Beverages	0.16						

Table 1: Summary statistics. HHI, DHHI, merging party share, average market size, and number of products are all averaged at the market (DMA-month) level within merger. Other departments include Frozen Food, General Merchandise, Non-Food Grocery, and Packaged Meat.

mergers have limited impacts on concentration: the median merger has a change in HHI (DHHI) less than 20, although about a third of mergers in our sample have a DHHI larger than 100. We also have a large variety of different mergers in the dataset. Variation in the market size of the mergers is large, and there is a fair amount of variation in the number of products per merger. Finally, the most common NielsenIQ “department” for our mergers is dry groceries.

We are unable to obtain reasonable demand estimates for some of the mergers in our dataset: Section III.B provides the criteria. Panel B shows the same summary statistics for the sample for which we have estimates, and the patterns are similar. The share of large DHHI merger is slightly smaller, and we have a smaller share of mergers in health and beauty products.

Table B.1 shows the characteristics we collect. Since they are varied across mergers, we are unable to provide detailed distributions for each of the potential characteristics. However, given the characteristics are used to capture substitution between products, we compute distance between these

	Acquirer	Target	Non-Merging Parties
A. All Characteristics			
Acquirer	1.56	1.80	1.99
Target	1.80	1.48	2.12
Non-Merging Parties	1.99	2.12	1.95
B. Price			
Acquirer	0.53	0.82	0.88
Target	0.82	0.61	1.00
Non-Merging Parties	0.88	1.00	0.83
C. Non-Price Characteristics			
Acquirer	1.27	1.38	1.57
Target	1.38	1.07	1.62
Non-Merging Parties	1.57	1.62	1.55

Table 2: Average Euclidean distance in normalized characteristics space between competing products. All characteristics are normalized into z -scores before computing distances. Averages are weighted by shares within markets and then unweighted across markets and mergers.

characteristics. To compute this distance in a uniform way across mergers, we conduct a principal component analysis of each set of characteristics, take the first two principal components, and we normalize these to z -scores. (In Section III.A, we note that procedure maps to the characteristics used for demand estimation.) We then compute the average Euclidean distance between competing products. Each panel of Table 2 shows this distance between different sets of products. Our first observation is that the diagonal of each matrix is smaller than the elements in corresponding rows and columns: the acquirers' products are more similar to their own products than to other firms'. Second, the distance between the acquirer and the target products is smaller than that between the non-merging parties' products and either the acquirers' or targets'. This provides some evidence that firms with similar products are more likely to merge, an observation that has not been made systematically in the literature. Moreover, to the extent that these characteristics are valued by consumers, they will feed into larger diversion between merging parties than between merging and non-merging parties.

III. Demand Estimation

III.A. Model

For each merger, we estimate a differentiated products demand system following Berry et al. (1995), which has become the standard model in industrial organization for demand for consumer packaged goods. We define a market to be a DMA-month pair and aggregate price and quantities to the

product level across all retailers and weeks within a DMA, weighing these prices and quantities by the relative share in each retailer-week. For each product j in DMA d at time t , we define the indirect utility to consumer i as

$$u_{ijdt} = \beta_i^0 + \alpha_i p_{jdt} + \beta_i' \mathbf{x}_j + \xi_j + \xi_d + \xi_{\text{quarter}(t)} + \xi_{jdt} + \epsilon_{ijdt}, \quad (1)$$

where p_{jdt} is the price of the product, $\mathbf{x}_j$ is a vector of product characteristics, $(\xi_j, \xi_d, \xi_{\text{quarter}(t)})$ are product, DMA, and quarter-of-year fixed effects, ξ_{jdt} is an unobserved shifter of utility common across all individuals, and ϵ_{ijdt} is a Type I extreme value shock that is iid across products. The utility of the outside option is $u_{i0dt} = \epsilon_{i0dt}$, another independent draw from a Type I extreme value distribution. We parameterize the vector of random coefficients as

$$\begin{aligned} \beta_i^0 &= \Pi^{\beta^0} D_i \\ \alpha_i &= \alpha + \Pi^\alpha D_i + \sigma^\alpha v_i^\alpha \\ \beta_{ki} &= \Pi^{\beta^k} D_i + \sigma^k v_i^k, \end{aligned} \quad (2)$$

where the $v_i^\times$ are independent standard normal draws, and D_i is a vector of demographics. We use θ to refer to all the structural parameters to be estimated, and this consists of α , all $\Pi^\times$, and all $\sigma^\times$.

The parameterization in (1) and (2) allows for both vertical differentiation across products and horizontal differentiation through the random coefficients. The random coefficients also allow for a departure from share-based substitution and allow for substitution towards products that are closer in characteristics space or in price. The demographic coefficient on the constant β_i^0 breaks share-based substitution to the outside option: we do not add a random coefficient to this term since we have found it difficult to estimate precisely with the data we have. Throughout our analysis, we focus on an indicator for whether household income is above \$100,000 as the demographic of interest, although in some specifications we set some $\Pi^\times$ to 0.

Papers that estimate demand at scale (Atalay et al., 2025; Döpper et al., 2025) do not use product characteristics to model substitution, likely due to both the difficulty in collecting this data and the difficulty in choosing characteristics to use in the demand system. As mentioned in Section II.C, we tackle the second problem by using the first two principal components of the set of characteristics we collected by hand, an approach used by Backus et al. (2021). A benefit of this approach in our setting is that it allows us to avoid arguably arbitrary decisions about the choice of characteristics for each market and limits researcher degrees of freedom. It also allows us to account for discretely coded characteristics in a sensible manner. The first two principal components do not always capture a large share of the variation in the characteristics: they explain more than 50% of the variation in characteristics in about half of the mergers (see Table B.1). However, this approach allows us to incorporate the most variation possible from the characteristics while respecting computational

limits. Of course, this approach also limits the interpretability of random coefficients themselves. Perhaps more importantly, it is not clear that distance in the space of hand-coded characteristics is the variation that is perceived by consumers when they assess whether products are substitutes: an approach such as in Magnolfi et al. (2025) that determines characteristics from embeddings of consumer surveys would be promising, but infeasible for historical data.

Finally, we must compute market sizes (or, equivalently, shares of the outside option). Some prior papers studying individual markets take market sizes as proportional to population, where the proportion is based on an upper bound for inside purchases devised from facts about the product market (Nevo, 2000; Backus et al., 2021). The diversity of product markets in our sample makes this approach more difficult to implement in our setting. Instead, we take a simpler approach by calibrating the market sizes in each DMA as 1.5 times the largest sales quantity in that DMA throughout the sample, such that the maximum share of the outside good is 1/3. This is consistent with the approach in Miller and Weinberg (2017) and Atalay et al. (2025).

III.B. Estimation

Demand estimation follows the algorithm of Berry et al. (1995). Given a guess of the parameters θ , we solve for values of mean utilities δ_{jdt} that make observed and predicted shares match. We then project δ_{jdt} onto linear components $(\alpha, \xi_j, \xi_d, \xi_{\text{quarter}(t)})$ with two-stage least squares, instrumenting for α , and recover $\xi_{jdt}(\theta)$. We then search for values of θ that minimize a GMM objective including both standard instrument-exclusion moments and micro-moments (Berry et al., 2004).

Identification requires at least four instruments: one for the mean price coefficient and one for each of the random coefficients on price and the first two principal components of characteristics. Our main class of instruments is the differentiation instruments formulated by Gandhi and Houde (2019). For each characteristic, we compute the quadratic distance between product j 's location in characteristic space and that of its competitors as $z_{jdt}^x = \sum_k (x_{jdt} - x_{kdt})^2$. Relevance comes from the notion that products that face fewer nearby competitors along a certain dimension of characteristics space tend to have higher prices, while exogeneity comes from the assumption that product characteristics are exogenous. These instruments are computed on the full set of characteristics before taking principal components. We then estimate a measure of predicted price given exogenous variables $\hat{p}_{jdt} = \mathbb{E}[p_{jdt} | x_{jdt}, z_{jdt}]$ using a linear fixed effects regression, controlling for DMA-product fixed effects and all previously computed differentiation instruments, and construct a differentiation instrument from the fitted values $z_{jdt}^{\hat{p}} = \sum_k (\hat{p}_{jdt} - \hat{p}_{kdt})^2$.

We also consider two potential alternative classes of instruments. First, for many mergers, we are able to identify inputs that have producer price indices in FRED, and we use the indices as instruments that exogenously shift costs. We interact these with package size because input costs should be expected to shift the prices of larger products by proportionately more. For a subset of

these mergers, only products sharing a certain characteristic are expected to use a particular input; in these cases, we also interact the input prices with an indicator for whether the product has that value of the characteristic. Second, as an easily generalizable measure of common cost shocks, we use Hausman instruments: the average price of product j in time t in DMAs bordering DMA d . Naturally, a concern with this strategy is that bordering DMAs may have correlated demand shocks. Nevertheless, we have decided to incorporate these instruments due to their ubiquity in the literature and the dearth of cost side instruments for some markets.

For the demand estimates that enter our simulations, we want to use the most relevant set of instruments while maintaining the plausibility of the exclusion restriction and avoiding weak instruments. To that end, we prefer the differentiation and input price instruments over Hausman instruments when possible, as Hausman instruments require relatively stronger assumptions to be valid, but must occasionally rely on the Hausman instruments when input prices and differentiation instruments are too weak. The following instrument selection procedure captures these goals. We start with differentiation instruments (which are used in all specifications), and add input prices if their partial F -statistic in the first-stage regression of prices on differentiation instruments, input price instruments, and fixed effects is at least 20.⁸ If the input prices are insufficiently predictive, then we use only the differentiation instruments if their first-stage F -statistic is at least 100. Finally, if the differentiation instruments, by themselves or with input prices, are too weak ($F < 100$), we rely on Hausman instruments and differentiation instruments. We run one-step GMM with the instruments resulting from this iterative procedure, then compute the feasible approximation to the optimal instruments (Chamberlain, 1987; Berry et al., 1995; Reynaert and Verboven, 2014) and run two-step GMM with these instruments. We take this set of results as successful if the objective converges within our prespecified tolerance, the probability of an individual having a positive price sensitivity is less than 5%, the median own-price elasticity is between -0.5 and -20 , and the average predicted post-merger change in price is less than 25%.⁹ If estimation is unsuccessful with the initial set of instruments, we try the other possible sets of instruments in the following order: differentiation with input prices, differentiation only, and differentiation with Hausman.¹⁰

We also attempt to use the richest utility specification possible. We start by attempting to estimate the demand system where demographics enter all the random coefficients in (2). If estimation is

⁸The input prices only vary temporally unless interacted with characteristics, and for some product markets we are unable to identify producer price indices for instruments that comprise a substantial share of input costs.

⁹In practice, the final constraint only rules out estimates predicting nonsensically large price changes. In practice, this typically comes from our cost forecasting procedure, described in Section IV.A, returning large changes in costs rather than demand estimates directly. We are currently working on further disciplining this procedure to avoid such issues.

¹⁰Results using different sets of instruments are correlated, although not perfectly: the pairwise correlations in estimated median own-price elasticities are 0.57 with just differentiation instruments vs. differentiation and input prices, 0.36 for just differentiation vs. differentiation and Hausman, and 0.35 for differentiation and input prices vs. differentiation and Hausman.

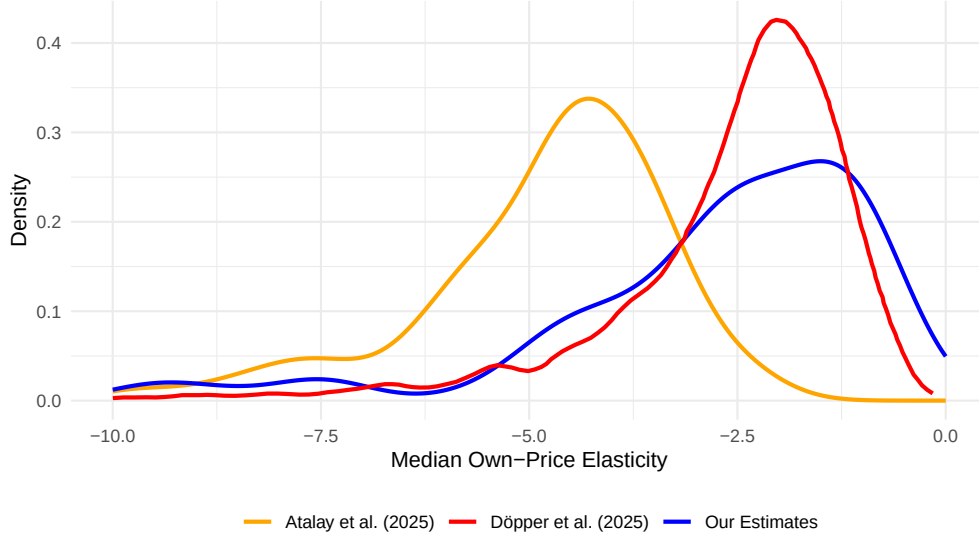


Figure 1: Comparison of our estimated elasticities with Atalay et al. (2025) and Döppler et al. (2025). Elasticities from Atalay et al. (2025) are unweighted 2010 elasticities.

unsuccessful, we consider a specification where demographics only interact with the constant and price coefficient. Finally, we consider a specification where demographics interact only with prices. To identify the demographic interactions, we use the Nielsen panel data to formulate micro-moments $\mathbb{E}[x_j|D_i]$, the average value of the characteristic conditional on the demographic (Berry et al., 2004; Conlon and Gortmaker, 2025).¹¹ In no case do we drop characteristics or random coefficients. To ensure we adhere to best practices, we use pyBLP (Conlon and Gortmaker, 2020) for estimation. Finally, to relieve concerns about local minima, we start estimation at two points: the result from nested logit using inside-outside nests, and the solution to the “FRAC” procedure of Salanié and Wolak (2022). If solutions from these initial points do not coincide, we restart estimation at a grid of starting points around these two values and pick the best solution. The instruments and specifications used for each merger in our estimation sample are listed in Table B.1.

III.C. Estimates

Figure 1 and Table 3 show the distribution of estimated median own-price elasticities across mergers. The median of this distribution is -2.46. The bulk of the distribution of median elasticities is between -1 and -5, with an interquartile range of [-3.93, -1.41]. However, there is a relatively sizable tail of more elastic estimates (some of which are truncated from the plot for readability); the average across all mergers is -3.39, with a 10th percentile value of -7.62.

As these demand elasticities are a key input into predictions from merger simulations, we

¹¹For demographic interactions with the constant, we use $\mathbb{E}[D_i|j > 0]$; that is, the average value of each demographic level given purchase of an inside good.

	Mean	St. Dev.	Quantiles				
			10%	25%	50%	75%	90%
Our Estimates	-3.39	3.09	-7.62	-3.93	-2.46	-1.41	-0.91
Atalay et al. (2025)	-4.88	1.60	-7.24	-5.55	-4.57	-3.84	-3.42
Döpper et al. (2025)	-2.13	1.12	-4.36	-3.11	-2.25	-1.65	-1.16

Table 3: Comparison of our estimated elasticities with Atalay et al. (2025) and Döpper et al. (2025). Elasticities from Atalay et al. (2025) are unweighted 2010 elasticities.

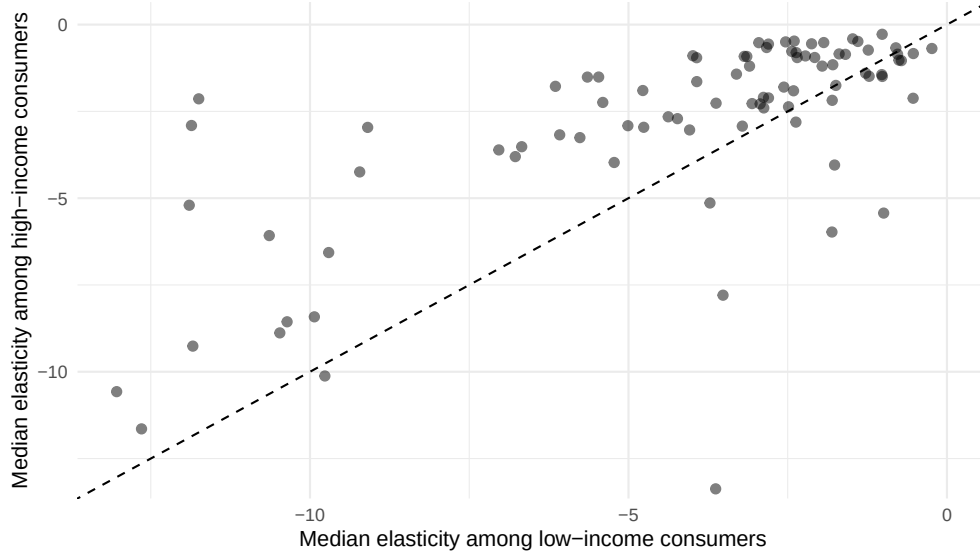


Figure 2: Median elasticities for low- and high-income consumers

perform a number of auxiliary checks on them. First, we compare these elasticities to others in the literature. Figure 1 and Table 3 report elasticity distributions estimated by Atalay et al. (2025) and Döpper et al. (2025). The main part of our distribution is more similar to that of Döpper et al. (2025), while Atalay et al. (2025)’s distribution is more elastic in the middle. However, the left tail of our distribution is more similar to that of Atalay et al. (2025). Table A.1 provides a point-by-point comparison of elasticities when possible: our estimates are generally less elastic than those in the literature, and there are some markets where our estimates lie far from other estimates. However, for most markets, our estimates are close to the (sometimes large) range documented in other work. These comparisons are not perfect given differences in the definition of products and markets across papers, but overall these exercises provide some evidence that our estimates are not out-of-line from the literature.

Second, we ask whether our demographic interactions are sensible. Work that estimates demand for consumer packaged goods typically finds that high-income consumers are generally less price sensitive than low-income consumers (Backus et al., 2021; Döpper et al., 2025). Figure 2 plots

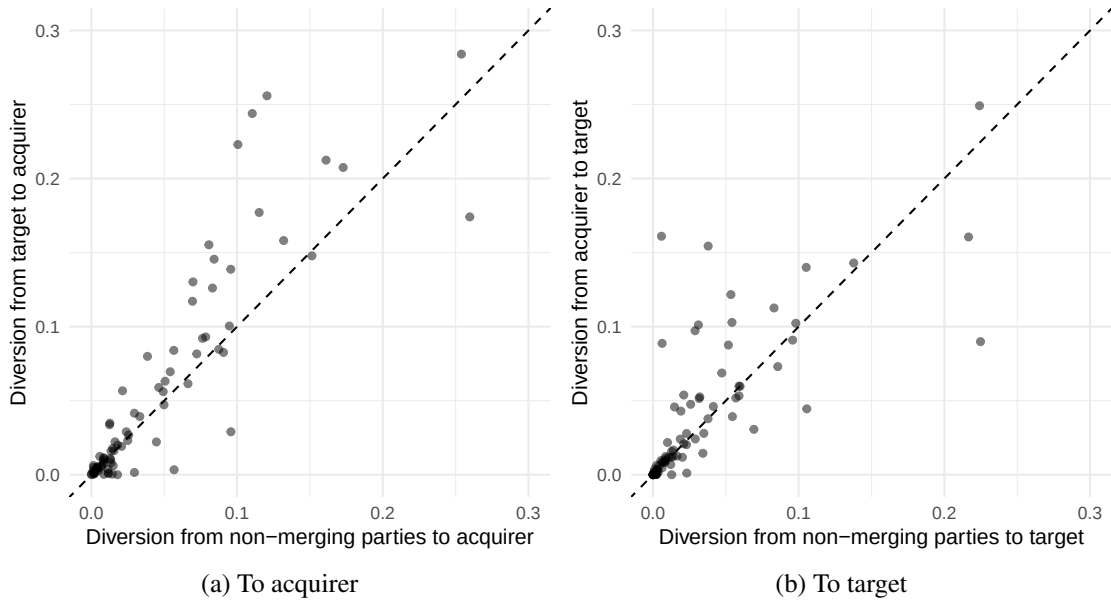


Figure 3: Comparison of diversion ratios between merging parties to diversion ratios across merging and non-merging parties

median own-price elasticities from each merger among low-income consumers on the horizontal axis and high-income consumers on the vertical axis. Consistent with the literature, we find that this is the case for the majority of our demand estimates as well. The median median elasticity is -3.12 for low-income consumers, as compared to -2.10 for high-income consumers.

Finally, we ask whether the demand system picks up on the relative product similarity of the acquirer and target products. Figure 3 plots the diversion ratios between the merging parties against the diversion ratio from the non-merging parties to the merging parties. We find greater diversion between the merging parties: on average, the target to acquirer diversion ratio is 61% (2.5 pp) larger than the non-merging party to acquirer diversion ratio, while the acquirer to target diversion ratio is 64% (0.3 pp) larger than the non-merging party to target diversion ratio. This suggests that the similarity in characteristics among the merging parties displayed in Table 2 are economically significant when analyzing potential merger effects. Moreover, we find that the mixed logit structure of the demand system does have an impact on these estimates: Figure A.1 plots the distribution of diversion ratios between merging parties and compares them to the share-based diversion we would expect from logit. We find that the diversion ratios are noticeably higher than logit in our estimates, especially in the upper tail, demonstrating that the demand system is estimating an important (and we think sensible) role for horizontal differentiation through the random coefficients.

In sum, our demand estimates appear reasonable. The own-price elasticity estimates, and the differences in price sensitivity between low- and high-income consumers, are in line with those from the literature. Finally, the demand systems pick up on the relative similarity in product

characteristics between the merging parties, manifesting in closer substitution patterns between the merging parties than their competitors. These results give us some comfort that our demand estimates will be a reliable input into the merger simulations.

IV. Evaluating Merger Simulations

We first ask how the predictions of merger simulations compare to actual realizations of prices. As we discuss in Section IV.A, we implement merger simulation as a forecasting procedure: we compare the predictions from the standard model to the realized prices after the merger, taking into account any observable changes in costs.¹² Given our sample size, we can first ask how predictions compare to the realizations on average. We then quantify the predictive power of these simulations at various levels—both cross-merger and within-merger. Throughout, we attempt to separate the predictive power of the simulation itself to that from observable changes to the system.

IV.A. Implementing the Merger Simulation

Under the Bertrand-Nash conduct assumption, prices $\mathbf{p}$ in a market solve

$$D(\mathbf{p}; \boldsymbol{\theta}, \boldsymbol{\xi}) + (\boldsymbol{\Omega} \odot \boldsymbol{\Delta}(\mathbf{p}))(\mathbf{p} - \mathbf{c}) = 0, \quad (3)$$

where $D(\mathbf{p}; \boldsymbol{\theta}, \boldsymbol{\xi})$ is the demand function (which we parameterize with the demand parameters $\boldsymbol{\theta}$ and demand shocks $\boldsymbol{\xi}$ to connect to our procedure), $\boldsymbol{\Delta}(\mathbf{p})$ is the transpose of the Jacobian of demand with respect to $\mathbf{p}$, $\mathbf{c}$ is the vector of marginal costs, and $\odot$ denotes elementwise multiplication. The ij entry of the ownership matrix $\boldsymbol{\Omega}$ is 1 if products i and j are owned by the same firm, and 0 otherwise.

The typical merger simulation uses the equilibrium conditions in (3) in two ways. First, if the researcher is willing to maintain the assumption that conduct in the pre-merger period is Bertrand-Nash, then (3) could be inverted to solve for costs $\mathbf{c}$ in every market. Second, the researcher can change the ownership matrix from $\boldsymbol{\Omega}$ to $\boldsymbol{\Omega}^{\text{post}}$, where $\Omega_{ij}^{\text{post}}$ is now set to 1 if products i and j are both owned by the merging parties. Holding demand and these recovered costs fixed, the researcher then solves (3) for the new equilibrium prices using this new ownership matrix.¹³ The difference in prices under the new and old ownership matrices is referred to as the unilateral effect of the merger (at a fixed demand and cost schedule).

¹²In this set of results we do not account for observable changes in demand. However, since the only such observable is the share of high income customers in the DMA, which is fairly stable over the time period of each merger, we do not anticipate this changing the results appreciably.

¹³While there is no closed-form solution for prices in this system, Conlon and Gortmaker (2020) recommends using an iteration proposed in Morrow and Skerlos (2011); we use this procedure, which is built into pyBLP.

In this section, we do not compute unilateral effects at a fixed demand and cost schedule but rather use this same methodology to compute a forecast p_{idtm}^{Pred} for the price of product i in DMA d in month t in merger m . Doing so requires us to take a stance on (i) the assortment of products in the market in this future period, (ii) demand shocks ξ the firms would face, and (iii) the marginal costs in that market. For (i), we take the set of products in each DMA in the final period before the merger: we are not trying to test models of entry and exit of existing products, and predicting the introduction of new products is beyond the scope of most models. For (ii), we take a simple approach of forecasting ξ in each DMA as the average of the ξ in the three months before the merger. For (iii), for each merger m , we first recover marginal costs c_{idtm} for each product-DMA-time combination in the pre-period. We then run a nonparametric regression of c_{idtm} on product characteristics x_{im} , cost shifters w_{idtm} , and time. Following Backus et al. (2021), we implement this using a random forest and obtain a cost function $h_m(x, w)$ and implied residuals $\omega_{idtm} \equiv c_{idtm} - h_m(x_{im}, w_{idtm})$. We then forecast c_{idtm} into post-merger periods t as $c_{idtm}^{\text{Pred}} = h_m(x_{im}, w_{idtm}) + \bar{\omega}_{idm}$, where $\bar{\omega}_{idm}$ is the average of ω_{idtm} in the three periods before the merger. That is, the main departure between our prediction and the price prediction that would be used to compute the unilateral effect is that we allow costs to change.

This procedure returns a prediction p_{idtm}^{Pred} for the price of product i in DMA d in month t in merger m . We will compare these with the actual prices p_{idtm} observed in the data. This departs from the typical approach in the literature of comparing the unilateral effect of the merger to some form of “causal” effects, estimated either through a simple difference (Peters, 2006; Björnerstedt and Verboven, 2016) or through a difference-in-difference (Weinberg, 2011; Weinberg and Hosken, 2013).¹⁴ We prefer the forecasting approach since it at least attempts to control for trends unrelated to the merger (changes in the price of input costs) in a way disciplined by the model. It also allows us to directly compare prices and predictions in the post-period rather than devising a strategy for an auxiliary causal estimate. However, this decision leads to two important and complementary limitations when interpreting our results. First, low accuracy or low predictive power of the forecast could be due to an inability to predict shocks unrelated to the merger rather than to misspecification of the model itself. We return to this issue in Section V.C. Second, the object of interest for an agency is the causal impact of the merger, not the total effect. We still think that evaluating the ability to predict the total effect is an interesting quantity that has not been studied at scale, but in Section IV.C, we have to be careful about evaluating the incremental power of merger simulations beyond what would be predicted from marginal costs changes alone.

¹⁴While testing merger simulations is not the main research question in Miller and Weinberg (2017), the approach (in their Figure 4) of comparing the time series of prices in the post-joint venture period under various models mirrors our approach here. Note also that using a simple difference as the causal effect of the merger and comparing to the unilateral effect of the merger from the simulation can be viewed as a version of our forecasting procedure under the assumption that nothing about the underlying market (e.g., demand and costs) changes.

In our analysis, we typically average p_{idtm}^{Pred} within merger-product-DMAs across units of time and obtain predictions p_{idm}^{Pred} . Our goal is to compare these prices to the actual prices p_{idm} in the post-period. Given the large variation in prices across products within a merger, and the lack of consistent units of prices across mergers, this comparison is most sensible when measured in (log) changes from the product’s original price p_{idm}^0 before the merger. Define $\Delta\text{Actual}_{idm} \equiv \log(p_{idm}) - \log(p_{idm}^0)$ and $\Delta\text{Predicted}_{idm} \equiv \log(p_{idm}^{\text{Pred}}) - \log(p_{idm}^0)$. We often aggregate these quantities further: the product-level change is computed using the average of $\Delta\text{Predicted}_{im}$ across DMAs d , the DMA-merger-level change $\Delta\text{Predicted}_{dm}$ is computed using the pre-merger share-weighted average of $\Delta\text{Predicted}_{idm}$ across i , and the merger-level change $\Delta\text{Predicted}_m$ is the average of $\Delta\text{Predicted}_{dm}$ across DMAs. Since our baseline merger simulation uses the product assortment from before the merger, we do not have p_{idtm}^{Pred} for any product that is not in the assortment before the merger. Thus, the averages we compute are restricted to products that existed both before and after the merger.

This process implicitly assumes that retail prices are set by the manufacturers of products. This assumption is common in a lot of work in consumer packaged goods, dating back to Nevo (2000), and underlies the above cited papers studying markups at scale as well. We anticipate that adding a retail markup will not change the results in the paper, as they will act as a tax on wholesale prices set by the manufacturer and the effect on marginal costs will be “undone” in the merger simulation, as in Miller and Weinberg (2017). However, in future drafts, we plan on calibrating a model with a retail sector to evaluate this claim quantitatively.

IV.B. Are Merger Simulations Well-Calibrated?

We first ask whether merger simulations are calibrated in expectation: is the expected price change from a merger equal to the prediction from the simulation? We separate this question into two parts. First, we estimate directional accuracy of the predictions. Second, we investigate the sign and magnitude of the error in the predictions.

Directional Accuracy. Directional accuracy simply asks whether mergers where we predict a larger price change have a larger realized price change in expectation. It is in principle a fairly undemanding requirement of merger simulations, but because it requires a sizable sample of mergers to test, it is not something that has been established before in the literature. Moreover, there are many reasons this could fail even if demand is well-estimated. For instance, mergers that would have large price effects might only be approved (or even proposed) if parties expect large cognizable synergies.¹⁵ A more extreme reason is that the theory that underlies merger simulations—that of unilateral price effects tied to substitution patterns—could be a poor representation of firms’ pricing

¹⁵The opposite correlation is also possible: firms may only be willing to merge in competitive markets if they expect to realize synergies.

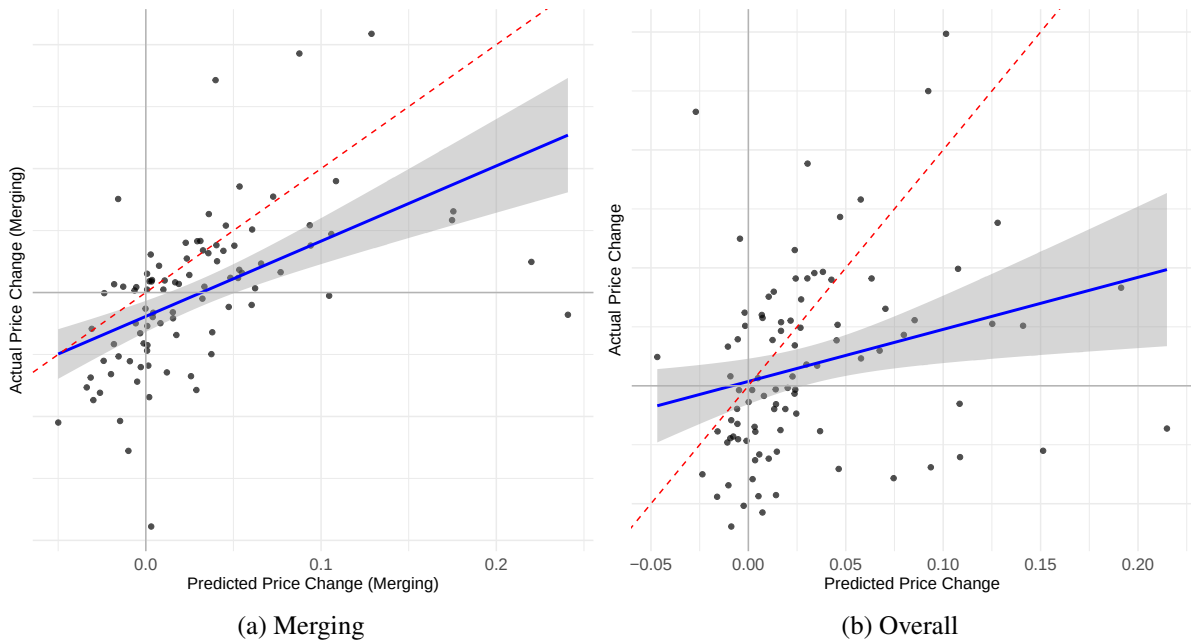


Figure 4: Scatter of actual versus predicted changes in prices, at the merger level, for (a) merging parties, and (b) all parties. The dotted red line indicates the 45° line, and the blue line is the line of best fit.

models in reality. Moreover, checking for directional accuracy at different levels is informative: even if parties change prices in a way that is on average consistent with predictions from merger simulations, effects in different geographies and products may be based on rules-of-thumb or uniform pricing rules (Escudero, 2018; DellaVigna and Gentzkow, 2019), mitigating the relation between predictions and realizations.

Figure 4 scatters ΔActual_m against $\Delta\text{Predicted}_m$ both for merging parties and overall. Both panels show that the predictions are correlated with actual price changes, although the predicted price changes do not change one-for-one with actual changes: the slope is 0.61 (s.e. 0.17) for merging price changes and 0.22 (s.e. 0.11) overall. Since the predicted price changes account for observable changes in cost shifters, they can be negative. However, this also means that the positive correlation is partly due to shifts in costs rather than the structure of the merger simulation itself.

To control for such changes, we run an extended version of the Mincer and Zarnowitz (1969) regression:

$$\Delta\text{Actual}_x = \alpha + \beta \cdot \Delta\text{Predicted}_x + \text{Controls} + \epsilon_x, \quad (4)$$

where the x denotes that the level of the regression changes in different specifications. We include the predicted change in marginal cost, as a proportion of the original price, as a control.¹⁶ When running regressions at the merger or the DMA level, we also control for HHI and DHHI—either

¹⁶When computing this quantity, we winsorize extreme outliers: those where the marginal cost changes by more than 5 times the initial price.

averaged across DMAs in the first case, or at the DMA-level in the second. Table 4 reports results of this analysis, with each column corresponding to a different level of analysis and each panel corresponding to a different set of products.

Column (1) runs the regression at the merger level. First, we find that predicted changes in marginal cost are positively correlated with actual price changes, and especially so for merging parties: this is expected and also provides some empirical validation for the estimated cost function. The coefficient is considerably larger for merging parties; we are cautious about interpreting this gap, but it could be suggestive of higher passthrough of merging parties, or these parties being more likely to re-optimize their prices around the time of a merger, including in response to cost shocks. Second, we find that price changes are positively correlated with DHHI: a 100-point increase in DHHI is associated with approximately a 0.4 pp larger price change. Even though the left-hand side does not purport to measure the causal effect of the merger, this correlation with DHHI is consistent with results from merger retrospectives (Kim and Singal, 1993; Focarelli and Panetta, 2003; Brot et al., 2024; Bhattacharya et al., 2025). The result germane to merger simulations is that predictions from merger simulations are positively correlated with actual prices even conditional on market structure and cost changes. The coefficient is 0.37 for merging parties, but approximately half as large overall and for non-merging parties.

Columns (2)–(4) of Table 4 rerun this regression at more granular levels of analysis: the DMA, product, and product-DMA. Patterns are generally similar. Predicted changes in marginal costs are generally significantly positively correlated with price changes, even at the DMA and product-DMA level—although the point estimate is negative and noisily estimated for merging parties at the product level. Price changes are positively correlated with DHHI, although the coefficient is smaller. Most importantly, merger predictions are positively correlated with actual price changes, and the point estimates are larger than in Column (1). At the product level, we cannot reject the Mincer-Zarnowitz benchmark of a slope of 1 at conventional significance levels, albeit in large part due to noise. The larger slopes at finer levels are consistent with aggregation dampening the signal in two ways: (i) within-merger heterogeneity can cancel out in share-weighted merger averages (so the regressor has less effective variation), and (ii) any noise in measurement can translate into greater attenuation when outcomes are compressed into a single merger-level statistic.

A complementary exercise asks whether merger simulations are directionally accurate within a given merger: do DMAs and products predicted to have large price increases have large price increases on average? This within-merger test is especially useful because it differences out merger-specific shocks that shift prices broadly (for example, integration-wide cost shocks, aggregate demand shifts, or merger-level implementation disruptions). Such results are also relevant for policy, as antitrust enforcement occurs not just at the merger level: agencies often require product divestitures, and they sometimes (but rarely) enforce regional divestitures as well.

	Cross-Merger				Within-Merger		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Merging							
Constant	-0.03** (0.01)	-0.02** (0.01)	-0.02*** (0.01)	-0.02*** (0.01)			
Predicted Δ Price	0.37*** (0.14)	0.56*** (0.13)	0.70** (0.29)	0.32*** (0.06)	0.58*** (0.08)	0.83** (0.41)	0.25*** (0.06)
Predicted Δ Marginal Cost	0.61** (0.26)	0.34** (0.13)	-0.54 (0.61)	0.43*** (0.09)	0.20** (0.09)	-0.72 (0.71)	0.47*** (0.10)
HHI	0.00 (0.04)	0.01 (0.03)			-0.06** (0.02)		
DHHI	0.43** (0.18)	0.15 (0.13)			-0.35*** (0.12)		
B. Overall							
Constant	0.01 (0.01)	0.00 (0.01)	-0.01** (0.00)	-0.02*** (0.00)			
Predicted Δ Price	0.19* (0.11)	0.33*** (0.10)	0.32*** (0.09)	0.47*** (0.04)	0.44*** (0.07)	0.31*** (0.10)	0.45*** (0.04)
Predicted Δ Marginal Cost	0.09 (0.07)	0.09** (0.04)	0.20* (0.11)	0.16*** (0.05)	0.09*** (0.03)	0.20 (0.14)	0.17*** (0.05)
HHI	-0.04 (0.03)	-0.02 (0.02)			0.01 (0.02)		
DHHI	0.41*** (0.13)	0.17** (0.09)			-0.20*** (0.05)		
C. Non-Merging							
Constant	0.01 (0.01)	0.00 (0.01)	-0.01** (0.00)	-0.01*** (0.00)			
Predicted Δ Price	0.20** (0.10)	0.38*** (0.10)	0.27*** (0.09)	0.51*** (0.05)	0.58*** (0.12)	0.25*** (0.08)	0.50*** (0.04)
Predicted Δ Marginal Cost	0.09 (0.05)	0.08** (0.04)	0.27*** (0.09)	0.12** (0.05)	0.08** (0.04)	0.30*** (0.10)	0.12** (0.05)
HHI	-0.06** (0.03)	-0.03 (0.02)			0.02 (0.02)		
DHHI	0.37*** (0.12)	0.11* (0.06)			-0.11 (0.07)		
Level	Merger	DMA	Product	P-DMA	DMA	Product	P-DMA
Merger FE	No	No	No	No	Yes	Yes	Yes

Table 4: Regressions of actual change in (log) price on predicted change in (log) price, at various levels. Each column runs the regression at a different level of observation (with “P-DMA” standing for product-DMA). Panels A and C focus on merging and non-merging price changes only, respectively. Panel B reports results for all price changes. Measures of market structure are not defined at the product level and are thus omitted from Columns (3), (4), (6), and (7).

Relative to the cross-merger analysis, two additional, conflicting forces affect our interpretation of the within-merger results. First, given geography-specific remedies are uncommon, it is unlikely that synergies at the DMA level are negatively correlated with effects from the merger due to selection into merger approval. This force would act to increase the correlation between predicted and actual price changes. On the other hand, there is evidence that price changes are not sensitive to local market structure due to uniform pricing rules at the retailer level (DellaVigna and Gentzkow, 2019), although national firms still seem sensitive to local cost shocks (Butters et al., 2022).

We re-estimate (4) at the DMA, product, and product–DMA levels and include merger fixed effects, so identification comes from deviations from each merger’s own mean predicted and realized price change. The results are reported in Columns (5)–(7) of Table 4. Generally, the patterns in the within-merger regressions are similar to those in the cross-merger regressions in Column (2)–(4), with quantitative differences being within statistical uncertainty.¹⁷ Despite the additional forces at play within merger, directional accuracy still remains a feature of these predictions.

Prediction Error. While the above analysis shows that larger price forecasts from merger simulations on average indicate larger realized price changes, the scatter in Figure 4 suggests that predictions from merger simulations are not unbiased on average. We study the prediction error

$$\text{Error}_{idm} \equiv \Delta \text{Predicted}_{idm} - \Delta \text{Actual}_{idm} = \log(p_{idm}^{\text{Pred}}) - \log(p_{idm}).$$

We then aggregate Error_{idm} to different levels using the same procedure as for $\Delta \text{Actual}_{idm}$.

Figure 5 plots the distribution of the merger-level prediction error Error_m , both overall and split by merging and non-merging parties. Panel A of Table 5 provides summary statistics of this distribution. Our main finding is that while there is a wide distribution of prediction errors, they are typically negative: merger simulations frequently overpredict price increases. The average prediction error for merging parties is -3.1%, and merger simulations overpredict in approximately three-quarters of mergers. The overprediction is smaller for overall price changes, with a mean of -2.2% and an error distribution that is shifted to the right. This provides a more consistent pattern in results than the limited literature on evaluating merger simulations has provided thus far: Weinberg (2011), Björnerstedt and Verboven (2016), and Miller and Weinberg (2017) document underprediction; Weinberg and Hosken (2013) study one merger with overprediction and one with underprediction, and the sign in Peters (2006) depends on the chosen demand system.

We check two potential sources of this prediction error that would be unrelated to merger simulations. The difference is not due to an initial prediction error that fades as firms adjust to the

¹⁷The one striking difference is that DHHI is now negatively correlated with realized price changes. Note that this is conditional on the predicted price change. Dropping the predicted price from the regression recovered positive, although statistically insignificant, coefficients on DHHI.

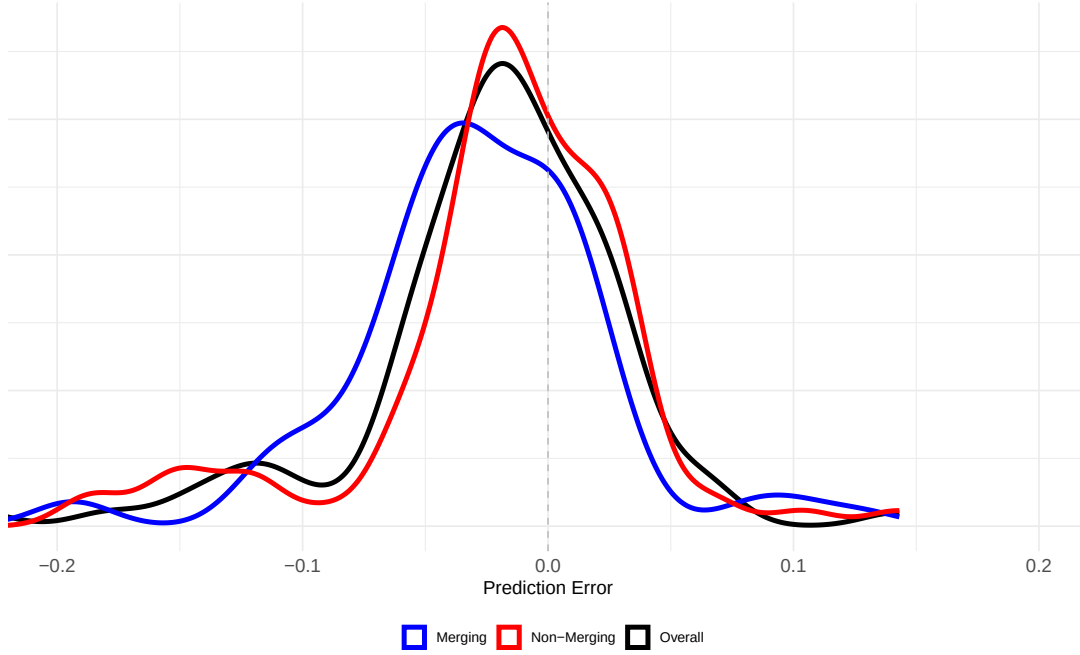


Figure 5: Distribution of prediction errors at the merger level

			Quantiles				
	Mean	St. Dev.	10%	25%	50%	75%	90%
A. Baseline							
Merging	-0.031	0.056	-0.089	-0.056	-0.030	0.002	0.018
Overall	-0.022	0.054	-0.080	-0.043	-0.018	0.010	0.027
Non-Merging	-0.019	0.055	-0.088	-0.034	-0.014	0.013	0.030
B. Using Post-Merger Demand and Assortment							
Merging	-0.037	0.077	-0.097	-0.066	-0.029	0.004	0.029
Overall	-0.020	0.098	-0.087	-0.041	-0.011	0.012	0.046
Non-Merging	-0.017	0.098	-0.087	-0.033	-0.007	0.015	0.041
C. Adding Partial Best Responses							
Merging	-0.034	0.064	-0.099	-0.064	-0.031	0.000	0.017
Overall	-0.021	0.096	-0.088	-0.039	-0.012	0.011	0.042
Non-Merging	-0.017	0.101	-0.089	-0.030	-0.008	0.014	0.040

Table 5: Summary statistics of prediction error (log actual price minus log predicted price) on market characteristics, at different levels of observation.

new equilibrium: the prediction error grows in magnitude over time (Figure A.2). We also compute the prediction error from running this procedure at a point when a merger did not occur. We conduct a “merger simulation” with no ownership changes at a date one year before the merger and predict prices for one year after the placebo date. We find that the prediction error for all parties is centered

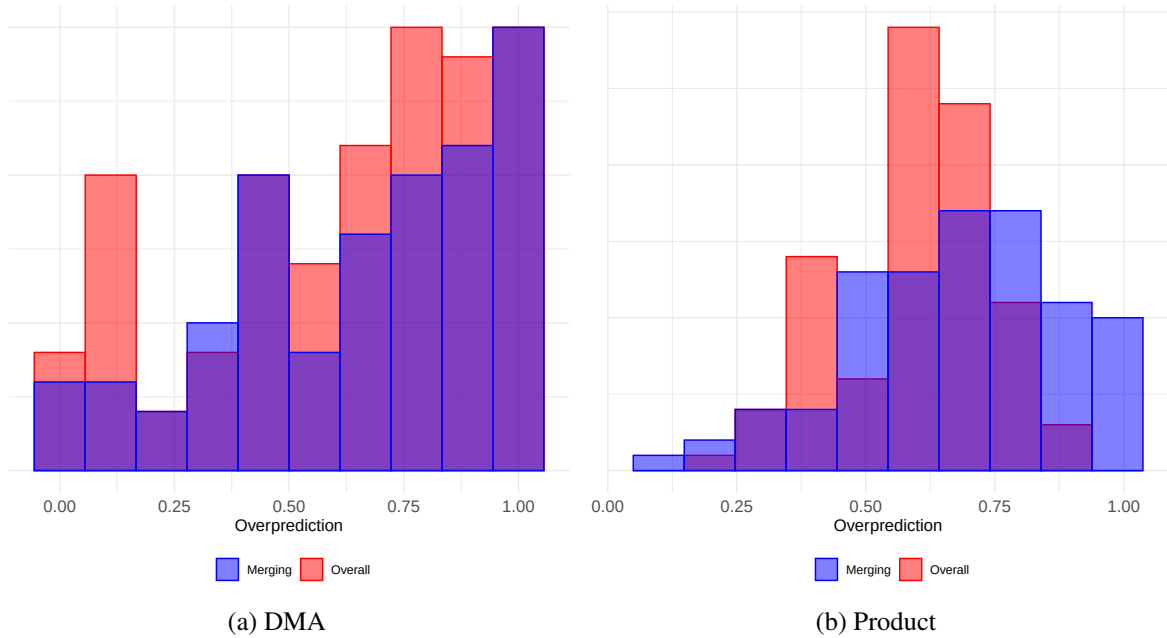


Figure 6: Histograms of the share of (a) DMAs or (b) products within a merger in which there is an overprediction of price increases

around zero (Figure A.3), providing evidence that the prediction error we observe is not due to secular trends in costs or demand that are not captured by our forecasting models.

Figure 6 plots the share of DMAs within each merger where the simulation overpredicts the price increase (Panel (a)) and the share of products within each merger where the simulation overpredicts the price increase (Panel (b)). The shape of these distribution is indicative of whether errors in predictions are clustered within merger. Consistent with the negative means in the price distribution, these distributions have more mass to the right of 0.5 than they do to the left. However, with the exception of the distribution for overall price changes averaged from products, the distributions show wide distribution: if overprediction were scattered randomly across mergers, we would expect the histograms in Figure 6 to be masspoints at the average probability of overprediction. About one-fourth of all mergers have a realized price increase lower than the predicted one in at least 90% of the DMAs (about one-fifth when studying product-level overprediction). Of course, this is still far from the extreme benchmark where overprediction is perfectly correlated within-merger.

While the clustering suggests merger-level misspecification that could lead to these prediction errors, we have not found merger-level observables that are predictive of the prediction error.¹⁸

¹⁸We have found that the magnitude of merging party prediction errors is negatively correlated with total revenue of the merger party at the DMA-level, both within- and across-merger. The effects range from -0.32 pp (s.e. 0.15) to -0.69 pp (s.e. 0.17) per log point for merging parties, depending on whether the effect is estimated within-merger and whether other DMA-level covariates (merging share or DHHI) are included as controls. We do not find a similar correlation when using prediction error itself. This could suggest departures from standard models of conduct in

	Cross-Merger				Within-Merger		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Merging							
Individually	0.265	0.245	0.043	0.089	0.187	0.032	0.079
Over Costs	0.164	0.110	0.077	0.019	0.067	0.081	0.009
Over Costs and Market Structure	0.105	0.086	—	—	0.076	—	—
B. Overall							
Individually	0.068	0.119	0.039	0.070	0.142	0.038	0.068
Over Costs	0.059	0.093	0.014	0.028	0.106	0.012	0.022
Over Costs and Market Structure	0.050	0.078	—	—	0.118	—	—
C. Non-Merging							
Individually	0.054	0.081	0.039	0.068	0.118	0.038	0.067
Over Costs	0.029	0.060	0.010	0.029	0.092	0.007	0.025
Over Costs and Market Structure	0.055	0.064	—	—	0.090	—	—
Level	Merger	DMA	Product	P-DMA	DMA	Product	P-DMA
Merger FE	No	No	No	No	Yes	Yes	Yes

Table 6: Partial R^2 of predicted change in prices over different characteristics of the merger

Table A.2 shows correlations in prediction error within-merger with DHHI. In Section V, we turn to estimating synergies and conduct at the merger level—even if they are not explained by merger-level observables at our disposal—and such within-merger patterns inform our estimates.

IV.C. Are Merger Simulations Predictive?

The analysis in Section IV.B is relevant for researchers interested in identifying systematic patterns the predictions from this common tool in industrial organization. However, it does not speak to the uncertainty in these predictions. Enforcement agencies would typically be interested in the predictive power of merger simulations since their decisions are implicitly based on loss functions that directly put weight on the “false positives” and “false negatives” from their decisions.

A simple first pass at forecast precision asks how much of the variation in price changes merger simulations can explain. Table 6 tackles this question by providing a series of partial R^2 values for various specifications. Each column corresponds to the associated column in Table 4. In each panel, the rows correspond to partial R^2 over the listed covariates. Panel A shows that the predicted change in price explains 27% of the variation in actual price changes, across mergers. Conditioning on the cost change, the prediction explains 17% of the residual variation, and it explains 11% of the variation left unexplained by costs and market structure. Panels B and C repeat this exercise with overall price changes and those of non-merging parties. Here, the values of R^2 are considerably

especially small markets, although we hesitate to speculate.

smaller: while predictions from merger simulations are positively correlated with the changes, the predictions explain approximately 5% of the residual variation in prices.

Column (2) of Table 6 shows that merger simulations remain reasonably predictive for merging parties at the DMA level. Predictive power improves for overall price changes and non-merging parties, relative to the cross-merger power. Comparing to Columns (2), (5), and (8) of Table 7, we see that the predictive power of merger simulations is typically larger at the DMA level than that of market structure by itself. Interestingly, simulations explain a noticeably lower share of the variation in price changes at the product or product-DMA level.

Column (5) of Table 6 shows the partial R^2 of price predictions beyond merger fixed effects. These values are slightly lower than ones in Column (2) for merging parties and in fact slightly higher overall and for non-merging parties. Thus, the predictive power of merger simulations at the DMA level is not coming purely from cross-merger predictive power: simulations do a similar job of predicting price changes across geographies within a market, a result that may be useful when deciding whether to use merger simulations to assess geographic remedies. The within-merger predictive power of product-level price changes or product-DMA level price changes is comparable to the cross-merger predictive power—and fairly low. Interestingly, the within-merger predictive power of merger simulations is much larger than that of structural presumptions by themselves, as seen by comparing Column (5) of Table 6 to Columns (3), (6), and (9) of Table 7.

We next ask about the accuracy of merger simulations in predicting the events that the price change after a merger is particularly large. Figure 7 plots receiver operator characteristic (ROC) curves for these events, for difference thresholds for price increases: the red curve, for instance, shows the predictive power for the event that the price change for merging parties is larger than 2.5%.¹⁹ Panels (a)–(c) show these curves at different levels of aggregation. We have a few observations. First, the predictions exhibit a moderate, but not especially high, amount of discrimination: the area under the curve typically lies between 0.65 and 0.73 overall and 0.70 to 0.80 for merging parties. Second, merger simulations do not show any particular bias towards avoiding false positives, say, and no part of the ROC curve is either unusually steep or unusually flat. Third, at the merger and DMA levels, the model shows a greater ability to discriminate especially large price changes. Importantly, however, these large price changes can also be due to marginal cost changes, and we show later that part of this power is coming from the improved predictive power of marginal cost changes at this point. We do not see any differences in discrimination ability across thresholds at the

¹⁹To plot an ROC curve, we fix ground truth events T_m —in our case, whether the price change after merger m is larger than 0%, 2.5%, or 5%. We then fix a threshold x and compute predictions $P_m(x)$ based on whether the merger simulation for merger m returns a price change larger than x . For each such x , we can compute the true positive rate (the share of the actual mergers with large price changes that the model correctly flags) against the false positive rate (share of actual mergers with low price changes that the model incorrectly predicts will have a high price change). We then vary x . A fully discriminating model will have an ROC curve that jumps immediately to (0, 1), while a fully random model has an ROC curve that coincides with the 45° line.

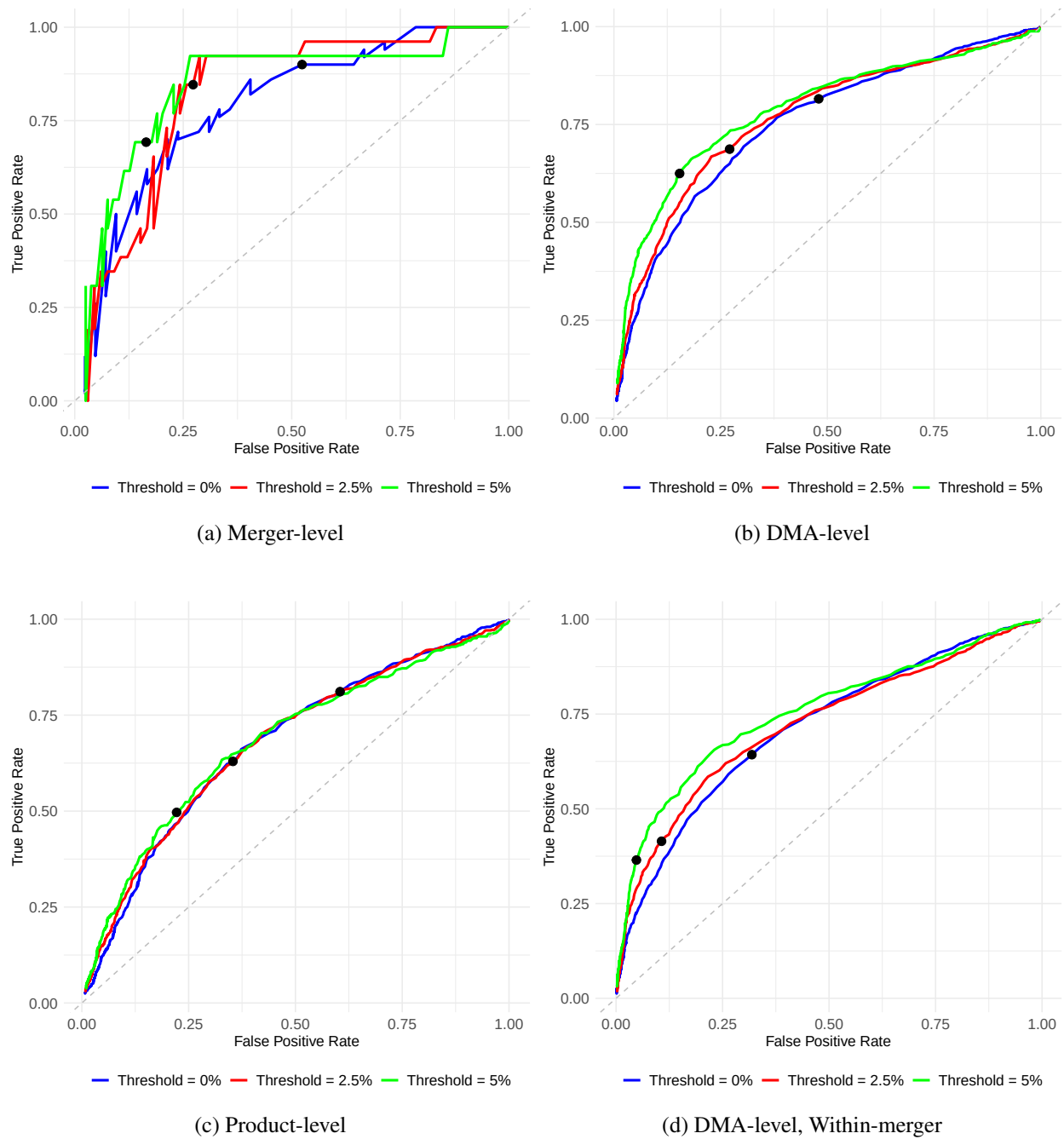


Figure 7: ROC curves for realization of price changes above a threshold, with forecasts from merger simulations as the prediction. Each panel shows the curves at a different level. The black dots in Panels (a)–(c) show the point on each ROC curve from the predicted value being larger than the same threshold used to compute the ROC curve.

product level. Finally, in Panel (d) we demean the DMA-level realized and predicted price effects within-merger to measure the prediction ability of the model within-merger. We see patterns fairly similar to the ones in Panel (b), although the ROC curves lie slightly closer to the 45° line.

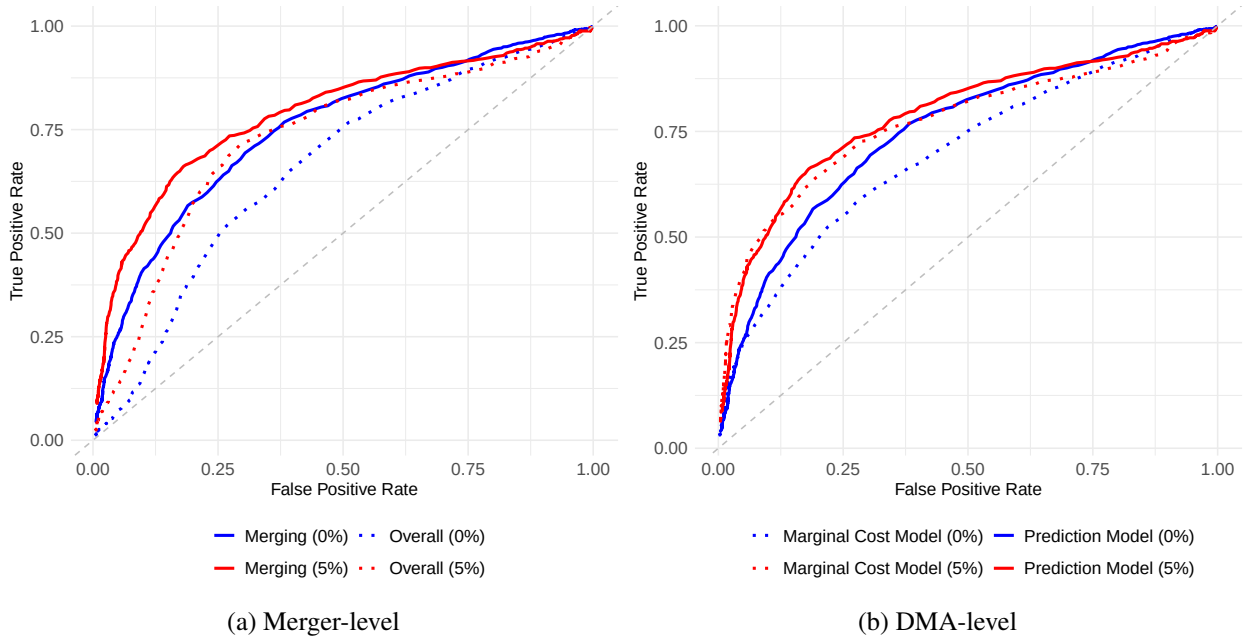


Figure 8: Comparisons of ROC curves for (a) merging and all parties and (b) predictions from simulations and those just from marginal cost changes.

ROC curves are invariant to monotone transformations of the prediction variable, but in our case, the prediction is generated for the same object that generates the signal. Accordingly, we distinguish with black dots the point on each ROC curve that corresponds to the prediction generated using the same cutoff as the threshold for that ROC curve. That is, the dot on the red line corresponds to using a prediction being larger than 2.5% as the binary prediction of the realized price change also exceeding 2.5%. These points (almost) all lie at a true positive rate larger than $2/3$: taking the prediction from the merger simulation at face value will flag the majority of true price increases. However, the false positive rate is also fairly high at these levels. This is consistent with predictions from simulations being biased upwards from the true value and provides a quantification of the types of mistakes an agency would make if they simply made decisions by taking the predictions from merger simulations at face value.

Panel (a) of Figure 8 compares the ROC curves for merging parties with those for predicting overall price changes. Consistent with lower R^2 for overall price changes, we find that the ROC curves for merging party price changes dominate the ones for overall price changes. Panel (b) compares the predictions to predictions based purely on marginal cost changes. For low thresholds, the merger simulation (in solid) clearly dominates simply using the marginal cost. However, the plot for the 5% threshold shows a caveat to our interpretation: while the AUC is large for the predictions from merger simulations, the ROC curves are much closer (and are not unambiguously ranked), and

	Merging			Overall			Non-Merging		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Individually	0.108	0.031	0.006	0.113	0.029	0.002	0.072	0.005	0.004
Over Costs	0.117	0.031	0.007	0.117	0.031	0.002	0.081	0.005	0.006
Over Costs and Predictions	0.055	0.004	0.017	0.109	0.015	0.015	0.105	0.009	0.004
Level	Merger	DMA	DMA	Merger	DMA	DMA	Merger	DMA	DMA
Merger FE	No	No	Yes	No	No	Yes	No	No	Yes

Table 7: Partial R^2 of market structure in prices over different characteristics of the merger

the merger simulations dominate the marginal cost model only starting around true positive rates of about 50%. Our interpretation is that merger simulations are not significantly better than just using changes in marginal costs in identifying the largest realized price changes, but they still have some incremental predictive content.

Our interpretation of these results is that merger simulations have some—but not especially large—predictive content for the total change in price after a merger. The low predictive content may come from an inability to forecast shocks. On the flip side, this predictive content is not merely capturing marginal cost changes: in both R^2 and ROC curves, merger simulations typically dominate using marginal costs by themselves. However, prior work (Bhattacharya et al., 2025) has also noted fairly low predictive power of other methods of predicting price changes, so it is important to benchmark these fairly low R^2 against such methods. Table 7 reports results from the exercise of computing the partial R^2 of the structural presumptions (HHI and DHHI) conditional on other metrics. Comparing Column (1) of Table 7 to Column (1) of Panel A of Table 6, we see that the predictive power for the structural presumption is significantly lower for merging parties than that of the simulation. However, Columns (4) and (7) of Table 7—compared to Panels B and C—show the opposite result for overall and non-merging parties: at the merger level, the predictive power of the structural presumptions is larger, although predictive power is generally low for all tools.

Another benchmark is how merger simulations compare to other methods used by practitioners. By modeling enforcement decisions, Bhattacharya et al. (2025) provide an estimate of the accuracy (share of classifications that coincide with the truth) of the agencies’ diligence process in determining whether mergers would raise or lower average price. They report an accuracy of 82%. The maximum accuracy merger simulations obtain for predicting the events that mergers raise prices by 0% or 2.5% is 63% and 70%, respectively (Figure A.4). As before, this comparison comes with the caveat that the object predicted in this paper is the total price change after the merger.²⁰

²⁰To this end, it is worth noting that the simulations achieve accuracies of 80–85% for predicting a price change above 5%, but we know from Figure 8(b) that predictive power of large price change is more reliant on marginal cost changes.

On net, our results show an advantage for merger simulations over simply using the structural predictions—especially for merging party price changes, and for within-merging price changes. However, the predictive power is still low, and it still likely falls short of the predictive power of the additional diligence than the enforcement agencies may be conducting.

V. Prediction Errors, Synergies, and Conduct

The results in Section IV are written for a user of merger simulations: we find that merger simulations are correlated with true price changes, typically underpredict these changes, and typically have more predictive power than other covariates used for price changes but nevertheless are of limited informativeness. This section evaluates these results by asking what merger simulations can elucidate about the structure of mergers. In particular, we investigate changes in demand, potential synergies, and changes in conduct.

V.A. Decomposing Prediction Errors

The merger simulations in Section IV are conducted assuming that neither demand nor product assortment changes after the merger. Both assumptions are unlikely to hold in practice. The merger may induce changes in advertising or deals with supermarkets that affect how products are sold, which would affect consumer demand, or the merging parties might have merged because they foresaw future changes to demand. Several retrospectives have identified product assortment changes after a merger (Atalay et al., 2024; Bhattacharya et al., 2025). With the post-merger data, we can estimate the share of the prediction error that is due to changes in assortment and demand.

To do so, we take the post-merger assortment in each market as fixed. We take the estimated demand parameters θ as fixed as well to avoid the (rather expensive) process of re-estimating demand. We then use the post-merger market shares and invert the demand system to recover the post-merger demand shocks ξ , which differ from the forecasted shocks used to simulate the merger in Section IV. We use the same marginal cost prediction as in Section IV and simulate post-merger prices as if demand were also known. We still use pre-merger shares to aggregate the prices so that we are comparing the same set of products.

Panel B of Table 5 shows the distribution of these prediction errors. Our interpretation is that changes in demand are not a main factor in explaining the prediction error. The mean prediction error for merging parties increases in magnitude to 3.6%, and the mean remains essentially unchanged for merging and non-merging parties. The distribution of prediction error becomes slightly wider in all cases. The limited effect of demand changes on prediction error mirrors the findings in Peters (2006), where this procedure was first proposed, and Weinberg and Hosken (2013).

Another explanation for our findings is that an error in predicting the costs of one of the parties

propagates to prediction errors for other parties. Merging parties may realize synergies, say, and non-merging parties in the market may be aware of these synergies and price lower than our model (without synergies) would expect. To evaluate this, we conduct an exercise where we compute the best response of non-merging parties as a whole to the observed post-merger price of the merging parties, and vice versa.²¹ We refer to these results as “partial best responses” in Panel C of Table 5. While some mergers are affected by this exercise, we find that the distribution of prediction error does not change appreciably relative to Panel B. This result reflects the low diversion estimated between merging and non-merging parties in Section III.C.

Our interpretation of these results collectively is that they point towards the important rule of what Peters (2006) calls the “supply side residual,” which captures everything not explained by demand. We investigate this in subsequent sections by quantifying synergies under the standard conduct assumption and then considering changes in conduct.²²

V.B. The Structure of Synergies Under Bertrand-Nash Conduct

Under the assumption of post-merger Bertrand-Nash conduct, the overprediction of price changes documented in Section V.A is attributable to marginal cost synergies. In this section, we take this conduct assumption as given and analyze the structure of these synergies, both to assess the plausibility that post-merger conduct is indeed Bertrand-Nash and as an object of interest in its own right. With estimates of post-merger demand, we can invert (3) using the realized prices at the realized product assortment and demand shocks ξ and obtain the implied marginal cost c_{idtm} . Then, we can define $\text{Synergy}_{idtm} \equiv (c_{idtm}^{\text{Pred}} - c_{idtm})/p_{idm}^0$, a quantity that is positive if marginal costs are below our prediction. We normalize this quantity by the original price of the good to bring units into the same scale and our synergies are thus measured as a proportion of price. As before, we aggregate this quantity over different units.

Figure 9 plots the distribution of implied synergies at the merger level, and Panel A of Table 9 provides summary statistics. The average synergy overall is 0.7% of the initial price. Merging parties realize an average synergy of 2.7%, with them realizing positive synergies on average in slightly fewer than three-quarters of mergers. Non-merging parties realize much smaller synergies on average, although the tails for non-merging parties are larger. It is in principle possible for non-merging parties to realize synergies from a merger: the merger may change bargaining with input suppliers, quantity changes may lead to changes in marginal costs if there are non-constant returns

²¹This exercise has similarities to Table 8 of Weinberg and Hosken (2013), which evaluates whether the post-merger first-order conditions are satisfied at the observed prices, to within statistical error.

²²The limited effect of partial best responses on non-merging party prices is also consistent with prices of non-merging parties responding more to prices of merging parties than one may expect from standard models. While this is not an explanation we pursue further in this draft, we see merit in doing so, as it mirrors an observation in Björnerstedt and Verboven (2016).

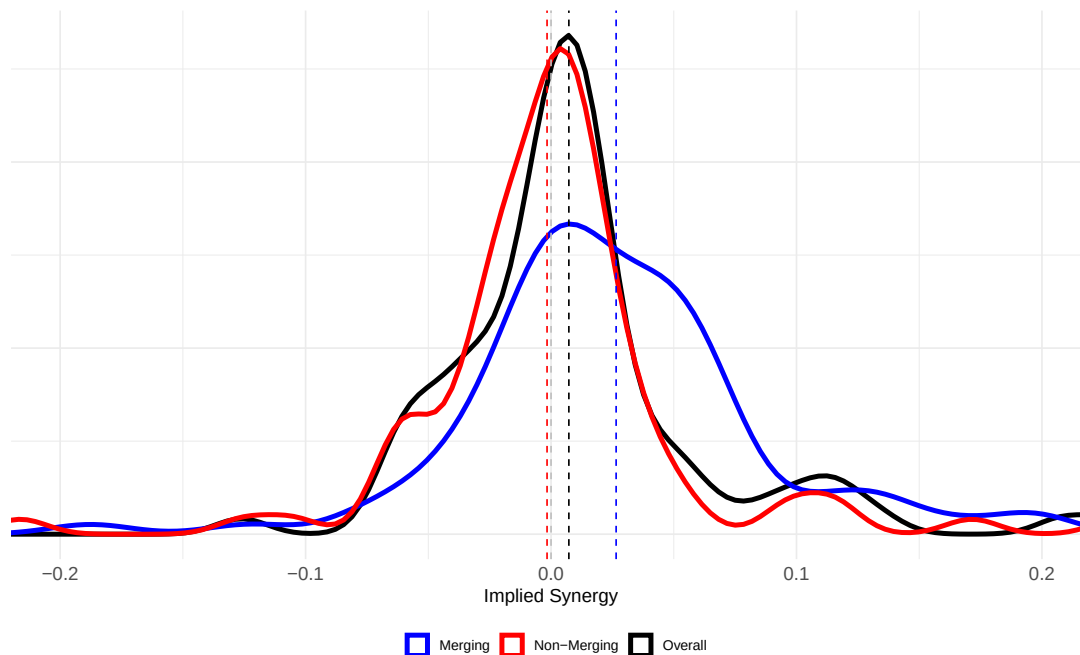


Figure 9: Distribution of implied synergies at the merger level, assuming Bertrand-Nash conduct

to scale (although we assume them away in our analysis in this paper), and the non-merging parties may be strategically induced into investing in cost reductions if their newly merged competitors realize them (Motta and Tarantino, 2021). However, these channels are at best less direct than the ones hypothesized for merging parties.²³

Figure 10 shows the distribution of synergies over time, split by merging and non-merging parties. The main observation is that synergies increase over time. Synergies for merging parties increase slightly over time, to a median of almost 3% 18 months after the merger. That synergies take time to materialize is consistent with plausible economic mechanisms for the source of these synergies—e.g., changes in distribution networks, change in production processes, or changes in negotiations with upstream suppliers. Prior work documenting synergies after mergers also note that synergies take time to materialize, sometimes even beyond the time period of this analysis.²⁴ At the same time, the box plots under “Months 1–6” document small immediate synergies for

²³To our knowledge, there is no direct evidence of synergies for non-merging parties in the literature: documenting that the price of non-merging parties could fall after a merger, which has been done in some papers cited elsewhere in this draft, could be attributed to strategic complementarities in pricing.

²⁴In the case of beer, Ashenfelter et al. (2015) uses distance to market as a measure of efficiencies and finds reductions about two years after the merger, stemming from the time it takes to reorganize production and distribution. Schmitt (2017) uses explicit measures of costs after hospital mergers and documents that about half the measured cost reduction happens 1–2 years after the merger but continues up to 4 years after the merger. Focarelli and Panetta (2003) and Sheen (2014) both use prices as a proxy for synergies: the former find price decreases in bank mergers starting approximately two years after consummation, and the latter finds a decrease in prices in consumer goods that speeds up after a year from the merger.

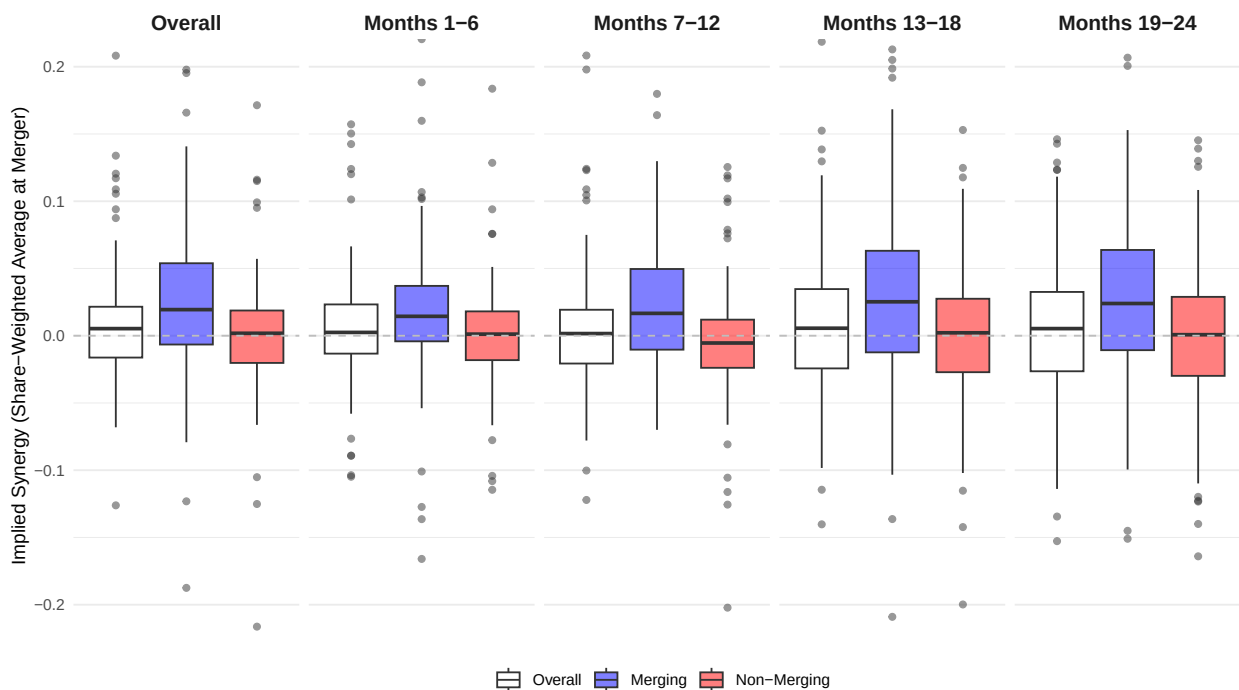


Figure 10: Distribution of implied synergies at the merger level, assuming Bertrand-Nash conduct, over time

merging parties, with a median synergy of almost 2% and approximately three-fourths of all mergers experiencing synergies. While immediate changes in costs are not implausible,²⁵ we would also see this pattern if firms do not take into account their market power immediately after the merger, perhaps if pricing must be renegotiated with supermarkets over many months. Alternatively, we would expect this pattern if the conduct model predicts that firms take “more advantage of” their market power than they actually do. Meanwhile, despite the potential mechanisms for post-merger cost changes for non-merging parties discussed above, the median non-merging synergy is essentially zero, both overall and in all time periods after the merger, although there is heterogeneity around this median.²⁶

The mechanisms hypothesized above—for merging parties but also for non-merging parties—imply a structure for the observed synergies. First, we would expect synergies to be correlated within-merger. Second, if synergies are due to efficiencies in distribution, we would expect them to

²⁵Demirer and Karaduman (2025) document efficiencies in power plants within a few months after the merger. While it is a priori unlikely that their mechanism applies to our setting of consumer packaged goods, it still provides evidence that efficiencies need not take years to realize.

²⁶There is a difference between the exercise that helps compute synergies and the ones conducted for the partial best responses in Section V.A. When computing synergies, we take prices of all rivals as fixed—including other non-merging parties. In the best-response exercise, we only fixed the prices of the merging parties and collectively changed the prices of the non-merging parties. The difference in these results may point to departures from Bertrand-Nash conduct for non-merging parties, although we have not explored that in this draft.

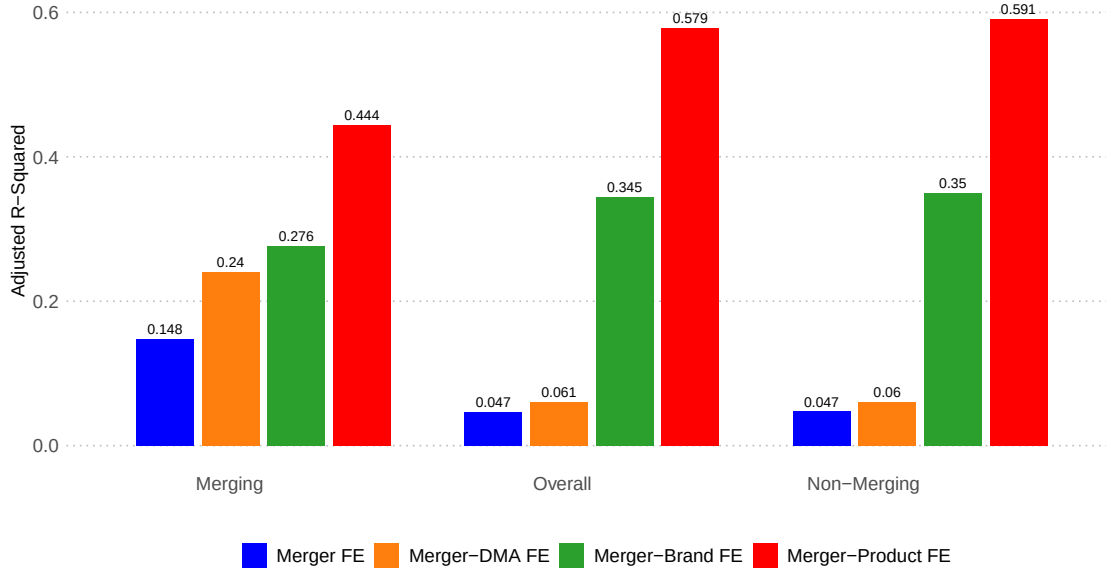


Figure 11: Adjusted R^2 values of regressions of fixed effects of various levels on implied synergies

be correlated within DMA across products. If they are due to improvements in production costs, perhaps because of unifying production lines, we would expect them to be correlated across DMAs within products or within similar types of products. We study this by regressing Synergy_{idm} on various sets of fixed effects and computing the adjusted R^2 of the fixed effects.

Figure 11 reports results from this exercise. For merging parties, about one-quarter of the variation in synergies across merger-DMA-products is explained by mergers themselves. About 24% is explained by merger-DMA fixed effects, suggesting that synergies are somewhat clustered within DMA after a merger, consistent with the importance of distribution efficiencies. We use brands as a proxy for whether products are similar: to the extent that products within the same brand umbrella benefit from the same production synergies, we would expect merger-brand fixed effects to have substantial predictive power over synergies. We see that they explain about 28% of the variation for merging parties. Finally, merger-product fixed effects explain about 44% of the the variation.²⁷ By comparison, mergers and merger-DMAs do not explain much of the variation in what we are calling synergies for non-merging parties, although more of the variation is explained by fixed effects related to products.

Which products realize the largest synergies? We take one specific approach to this question. We first compute Unilateral_{idtm} as the unilateral effect of the merger for product i in DMA d in

²⁷Of course, these mechanisms need not be the only reasons that implied synergies are correlated within DMAs or within product lines. However, it would have been more difficult to justify them as synergies if we saw very little correlation within these groups.

	(1)	(2)	(3)	(4)	(5)
Merging	1.52** (0.59)	1.49*** (0.27)	1.48*** (0.24)	0.88*** (0.20)	0.66*** (0.12)
Overall	2.32** (0.91)	1.90*** (0.30)	1.71*** (0.20)	0.97*** (0.19)	0.89*** (0.18)
Level	Merger	DMA	DMA	Product	Product
Merger FE	No	No	Yes	No	Yes

Table 8: Regression of implied synergies on unilateral effects, at various levels. Each cell corresponds to the coefficient on the unilateral effect in a separate regression. Results in the first row restrict to merging parties, and results in the second row use all parties. Standard errors are heteroskedasticity-robust in Column (1) and clustered at the merger level in all other columns.

post-merger period t and merger m —at the forecasted cost c_{dtm}^{Pred} and the recovered demand shocks. We aggregate as usual and regress Synergy_x on Unilateral_x at various levels, with and without merger fixed effects. The results in Table 8 show that synergies and unilateral effects are typically significantly positively correlated overall and for merging parties, for a variety of different levels.

There are reasons to believe that this correlation could be positive for merging parties. At the merger level, it may be that selection into merging is such that a merger with large unilateral effect is only allowed through if there are cognizable synergies. Within a merger across products, the products with the largest unilateral effects are closer substitutes for consumers, and this may be due to them being similar on dimensions that allow for production processes to be harmonized between them; this could lead to product-specific synergies for products with large unilateral effects. DMAs where one expects large effects are typically ones with a large presence of both merging parties: the merged entity may have an incentive to relocate distribution closer to these markets. However, another interpretation is that these potential mechanisms are somewhat far-fetched and that departures from the Bertrand-Nash conduct assumption are the explanation for these results. We turn to this in Section V.C.

V.C. Testing the Conduct Assumption

An alternate explanation for the prediction error is that the Bertrand-Nash assumption is incorrect. Given that overprediction is the modal finding, we focus on models that would deliver a lower price increase due to the merger than Bertrand-Nash. In particular, we focus on a model of conduct in which merging firms internalize a fraction $\tau \in [0, 1]$ of their merging partners’ profits in the post-merger period. With $\tau = 1$, this model coincides with Bertrand-Nash, and at $\tau = 0$ the model captures that the two units of merged entity continue to price separately. Intermediate values of τ capture a trade-off between a division’s own profits and its new partner’s profits that is less than one-for-one: similar models are commonly used in the literature (Crawford et al., 2018; Backus

et al., 2021), and in our setting this may come from, for instance, managerial incentives that depend both for firm profits and for their own division's profits.

We estimate τ by building off the observation in Table 8 and the associated discussion. In particular, our test is built upon detecting correlation between differences in merger-induced markup changes across models and unobservable marginal cost synergies, with the implicit maintained assumption that synergies ought to be uncorrelated with factors that change markups. We maintain throughout that conduct in the pre-merger period was Bertrand-Nash. In what follows, we estimate τ at the merger level but suppress the subscript m for notational simplicity.

Under our conduct assumption, the first-order conditions of the firms' price-setting problems imply

$$c_{idt} = p_{idt} - \eta_{idt}(\tau), \quad (5)$$

where c_{idt} is the marginal cost, p_{idt} is the price, and $\eta_{idt}(\tau)$ is the markup for product i in DMA d in month t under a particular model of conduct parameterized by τ . Note that this expression holds for both the pre- and post-merger periods, and we impose that $\tau = 0$ prior to the merger. We specify the change in marginal costs between the period immediately preceding the merger ($t = 0$) and a post-merger period t as

$$c_{idt} - c_{id0} = \Delta h(x_{idt}, w_{idt}) + \Delta \omega_{idt}, \text{ where } \mathbb{E}[\Delta \omega_{idt} | \mathbf{z}_{dt}] = 0. \quad (6)$$

In the above, x_{idt} captures product characteristics that affect both demand and marginal costs, w_{idt} capture variables that affect marginal cost and are excluded from demand, and $\mathbf{z}_{dt}$ is the full set of exogenous product characteristics, including those that affect both demand and marginal costs, those that affect only demand, and those that affect only marginal costs. We flexibly model the change in marginal costs through $\Delta h(\cdot)$. This function captures changes in the mapping from product characteristics and cost shifters to marginal costs between the pre- and post-merger periods, which could stem from factors such as improvements in the efficiency of input usage.

The conditions (5) and (6) together imply

$$(p_{idt} - p_{id0}) - (\eta_{idt}(\tau) - \eta_{id0}(0)) = \Delta h(x_{idt}, w_{idt}) + \Delta \omega_{idt}. \quad (7)$$

In (7), prices are observed, as are markups, given a value of τ and the estimated demand system. We can therefore treat the left-hand side of the expression as data. The model is then defined by the conditional moment restriction described above, $\mathbb{E}[\Delta \omega_{idt} | \mathbf{z}_{dt}] = 0$. However, in practice, we weaken this restriction and instead use $\mathbb{E}[\Delta \omega'_{idt} A(\mathbf{z}_{dt})] = 0$, where we choose $A(\mathbf{z}_{dt})$ in an analogous way to Backus et al. (2021), who motivate this choice by noting that the goal is to choose a function that provides us with the power to distinguish among the alternative models of conduct. Specifically, we

choose $A(\mathbf{z}_{dt}) = \mathbb{E} \left[\frac{\partial \Delta \omega_{idt}}{\partial \tau} | \mathbf{z}_{dt} \right] = \mathbb{E} \left[\frac{\partial \eta_{idt}(\tau)}{\partial \tau} | \mathbf{z}_{dt} \right]$, which corresponds to the feasible approximation to the Chamberlain (1987) optimal instrument.

For each merger, we estimate τ by GMM under the single moment restriction $\mathbb{E}[\Delta \omega'_{idt} A(\mathbf{z}_{dt})] = 0$. For each candidate τ , this procedure requires three steps. First, we estimate the function that determines changes in marginal costs and recover $\hat{\Delta \omega}_{idt}$. Second, we estimate $\frac{\partial \eta_{idt}(\tau)}{\partial \tau}$ as a function of $\mathbf{z}_{dt}$ and obtain the fitted values, $\hat{g}(\mathbf{z}_{dt})$. Backus et al. (2021) highlight the need to flexibly model both of these functions so, following their suggestion, we fit random forests. Finally, we compute the scalar moment

$$Q(\tau) = \left(\frac{1}{n} \sum \Delta \hat{\omega}_{idt} \hat{g}(\mathbf{z}_{dt}) \right)^2, \quad (8)$$

and we choose τ to minimize (8).

In addition to x_{idt} and w_{idt} , there are a number of classes of instruments that can enter $\mathbf{z}_{dt}$. These instruments must be excluded from marginal costs and explain differences in merger-induced changes in markups across different models of conduct (i.e., across different values of τ). In our case, we rely on Gandhi-Houde instruments, which leverage the intuition that merging-party products with few close substitutes in characteristics space would experience larger changes in markups were the parties to internalize pricing spillovers following the merger. We specify x_{idt} to include the first two principal components of product characteristics, package size, and month-of-year dummies, and w_{idt} to include input costs or Hausman instruments, depending on the selected specification.

Before moving to the results, we compare our approach with others in the literature. Backus et al. (2021) relies upon detecting correlation between markup differences across models and marginal cost shocks. Instead of running this procedure purely using post-period data, we choose to exploit the merger itself and difference across the pre- and post-merger periods. We view this as an advantage in our setting. By differencing, our test absorbs unobserved cost shocks that are persistent across time and instead relies on the orthogonality of cost changes and excluded demand shifters. Of course, it comes with a different assumption that changes to markups not explained by product-level factors are orthogonal to synergies, which is not directly comparable to assumption used by Backus et al. (2021).²⁸

This approach relates to the one proposed in Adão et al. (2025). Adão et al. (2025) studies tests of quantitative trade models and proposes a test for model-based predictions of policy effects using the goodness-of-fit measure: $\mathbb{E}[z(\Delta y - \Delta x)]$, where Δy is the observed change in outcome following the policy and Δx is the model-based prediction of the policy effect. In our setting, we

²⁸We should note that if one is concerned about synergies being correlated with factors that determine unilateral effects, then the exclusion restriction used in Backus et al. (2021) also would not hold—as the residual from the regression of costs in the post-period on cost shifters and product characteristics would incorporate the synergy and thus be correlated with the instruments they use.

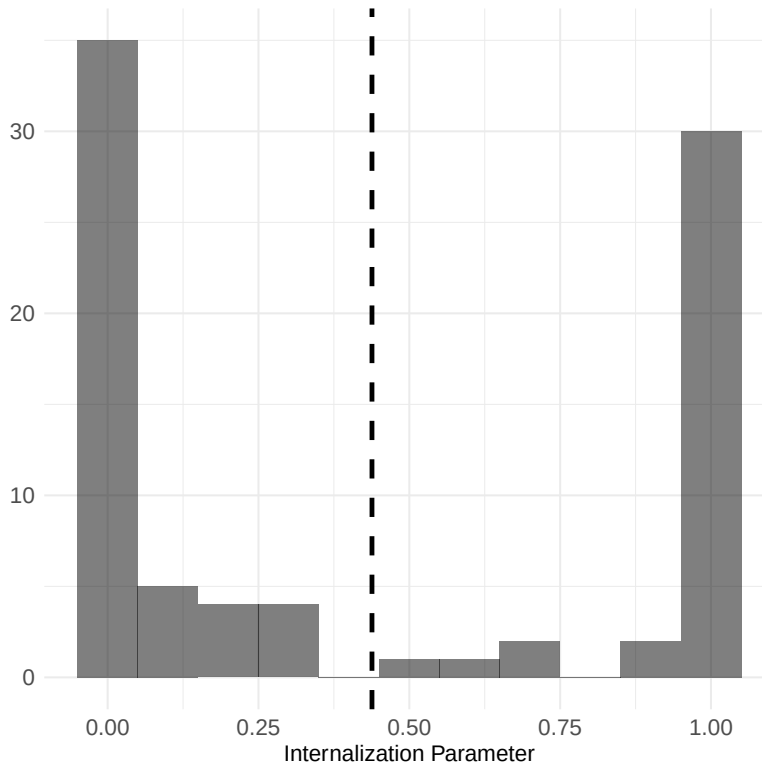


Figure 12: Estimates of internalization coefficients

could implement an analogous test, imposing that price changes following the merger are orthogonal to our set of instruments. However, the required structural assumption is that the instrument must be orthogonal to both changes in markups and changes in synergies.²⁹

Figure 12 shows results of internalization coefficients. We find an average internalization coefficient slightly below 0.5, with substantial bunching at the extreme values of 1 (Bertrand-Nash) or 0 (no internalization). We find that the internalization coefficient is significantly lower in mergers with larger DHHI and larger merging party share (see Table A.4). We also find that the internalization coefficient is slightly smaller in markets where merging parties are larger in terms of total revenue (a result that has a p -value around 0.12). In mergers with larger DHHI or market share for merging parties, we typically find that the correlation between the synergy and the unilateral effect is stronger under Bertrand-Nash—and the estimation procedure suggests that larger departures from Bertrand-Nash are necessary to rationalize the moment condition.

These results (while still preliminary) are somewhat at odds with our priors. One may expect that in larger markets, firms have more to gain from exploiting their newfound market power and would thus have an incentive to ensure that managerial frictions that could hinder pricing are minimized.

²⁹It is in this sense that the fact that we are unable to capture all changes in the environment before vs. after the merger, which we discuss at the end of Section IV.A, does not hinder our ability to test the model.

	Mean	St. Dev.	Quantiles				
			10%	25%	50%	75%	90%
A. Bertrand-Nash							
Merging	0.027	0.097	-0.034	-0.007	0.019	0.054	0.098
Overall	0.007	0.142	-0.049	-0.016	0.005	0.022	0.093
Non-Merging	-0.002	0.140	-0.056	-0.020	0.002	0.019	0.056
B. Partial Internalization							
Merging	0.021	0.066	-0.038	-0.009	0.028	0.045	0.076
Overall	0.033	0.074	-0.037	-0.004	0.031	0.051	0.101
Non-Merging	0.037	0.086	-0.035	-0.004	0.029	0.052	0.109

Table 9: Summary statistics for synergies

Markets where merging parties have a larger share are also likely to be more salient, and the merged entity may spend more time choosing new prices in these markets. On the other hand, markets where both merging parties are large before the merger may be the ones where there are separate (and institutionally important) pricing divisions already in charge of these products, and uniting them may be a more difficult task post-merger.³⁰ We are in the process of devising ways to extract information about the organizational structure changes from these mergers from SEC filings, and this auxiliary information could shed light on the plausibility of these explanations. However, for now we caution against interpretation that mergers are inherently less harmful for consumer surplus than the standard Bertrand-Nash model would predict, as we have not provided a structural reason why τ must remain below 1 in the post-merger period.

A positive interpretation of these results is that it allows us to reassess our estimates of synergies. Table 9 shows summary statistics of the distribution of implied synergies assuming partial internalization in Panel B. First, we find that the mean synergy for merging parties drops from 2.7% of price to 2.1% of price. This is driven by a drop in the right tail of estimated synergies: the modified conduct model attributes these overpredictions to departures from Bertrand-Nash conduct than to large reductions in marginal costs. At the same time, we see a slight increase in the median synergy: this is not because we predict significantly larger synergies for merging parties in any mergers but rather because some mergers where we predicted large synergies are now ones where we predict

³⁰Another explanation is that in these markets, the pre-merger conduct may be already be less competitive than Bertrand-Nash. We are currently ruling this explanation out in our analysis.

much smaller ones.³¹

V.D. Putting It All Together

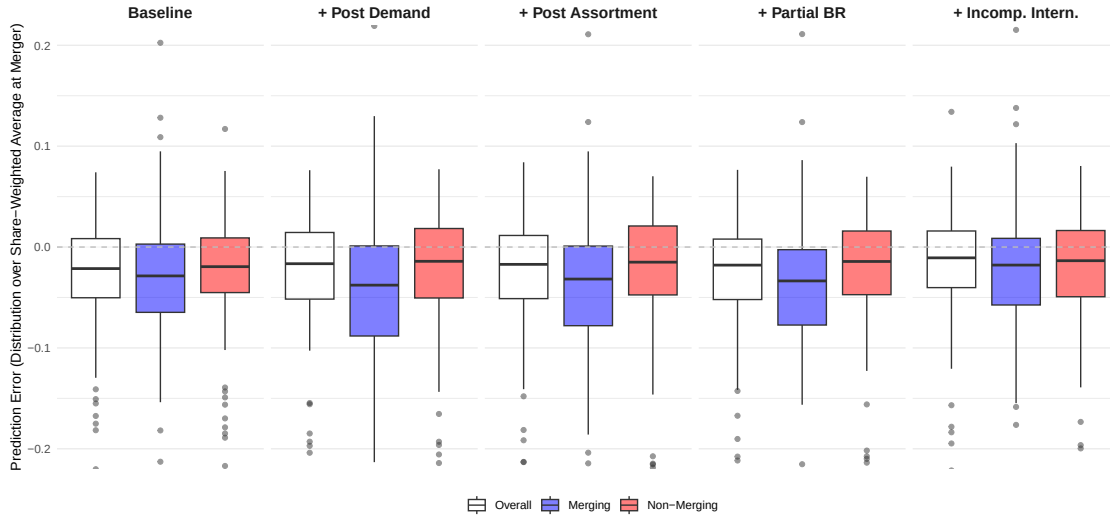
Figure 13 shows a decomposition of the distribution of prediction errors. The distributions under “Baseline” are box plot versions of Figure 5 (although we drop the year immediately following the merger) to allow prediction errors to stabilize). Adding in demand shocks and changes in product assortment has somewhat limited effects on this distribution, as does allowing for partial best responses. The largest changes come from allowing for incomplete internalization between the merging parties. Doing so reduces not just the median prediction error for merging parties but also the tail of the most extreme prediction errors. In fact, for mergers with DHHI larger than 100, allowing for incomplete internalization reduces the median prediction error to 0. Prediction errors remain for non-merging parties even after accounting for incomplete internalization, and the model attributes them to synergies. Overall, these exercises suggest a role for both synergies and departures from Bertrand-Nash in explaining the efficacy of merger simulations.

VI. Conclusion

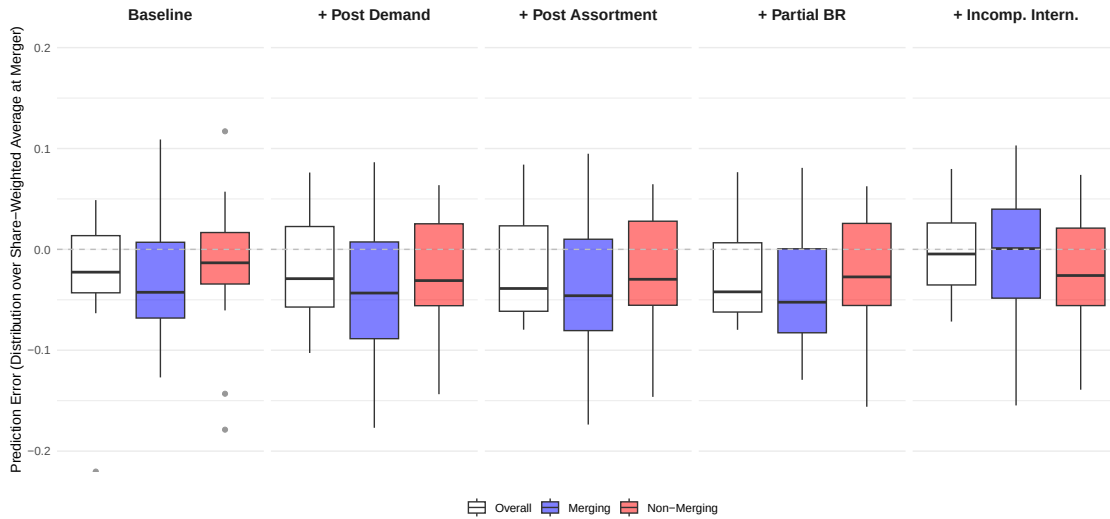
By comparing the predictions from a large set of merger simulation to the realizations from actual mergers, this paper identifies systematic properties of these predictions. Predictions and realizations are positively correlated. Merger simulations have more predictive power than the structural presumptions and add predictive power beyond simply using observable changes to costs, but their predictive power is still limited. Moreover, merger simulations systematically overpredict increases in price. We estimate average synergies of 2–3% of initial price for merging parties and also find systematic departures from Bertrand-Nash pricing: these are about equally important in explaining the departure of the realizations from the predictions of merger simulations, while changes to demand after the merger are much less important.

We think these results cast merger simulation in a moderately positive light. While our finding that merger simulation frequently overpredicts price increases cautions against naively taking the output of merger simulation as unbiased predictions, the additional predictive power merger simulation provides over market structure suggests that it can serve as an effective screen for mergers that are likely to increase prices. Further, the finding that half of the prediction error is due to

³¹The most salient difference, however, is that if merging parties are partially internalizing each others’ profits, the model implies that the *non-merging* parties realize rather large synergies. We are currently investigating the source of this discrepancy. Even if imperfect internalization lowers the overprediction for merging parties, thus leading to synergies being a more likely explanation for the overprediction for non-merging parties, prior results in this paper would not have suggested to us that synergies ought to be this large. Moreover, it is difficult to come up with mechanisms that would suggest that non-merging parties should have larger synergies than merging parties.



(a) Overall



(b) DHHI > 100

Figure 13: Decomposition of the prediction error for (a) all mergers and (b) high DHHI mergers. Distributions focus on the 12-24 months after the merger.

synergies suggests that performance can be improved even more by agencies with market-specific foreknowledge of cost improvements. However, we caution against overinterpreting this last result: these synergies are fundamentally a residual unexplainable by the family of conduct models we consider, meaning that any other departures from Bertrand-Nash conduct post-merger (or any departures at all pre-merger) would enter into synergies.

In ongoing work, we are studying the sensitivity of these results to the demand estimates themselves and asking how different the demand estimates need to be to ours to reduce the prediction

error. We are also comparing alternate moment conditions to test for conduct in the post-period.

References

- ADÃO, R., A. COSTINOT, AND D. DONALDSON (2025): “Putting Quantitative Models to the Test: An Application to the US-China Trade War,” *Quarterly Journal of Economics*, 140, 1471–1524.
- ANGRIST, J. D. AND J.-S. PISCHKE (2010): “The Credibility Revolution in Empirical Economics: How Better Research Design is Taking the Con out of Econometrics,” *Journal of Economic Perspectives*, 24, 3–30.
- ASHENFELTER, O. C., D. S. HOSKEN, AND M. C. WEINBERG (2015): “Efficiencies Brewed: Pricing and Consolidation in the US Beer Industry,” *RAND Journal of Economics*, 46, 328–361.
- ASKER, J. (2016): “Diagnosing Foreclosure Due to Exclusive Dealing,” *Journal of Industrial Economics*, 64, 375–410.
- ASKER, J. AND V. NOCKE (2021): “Collusion, Mergers, and Related Antitrust Issues,” in *Handbook of Industrial Organization*, Elsevier, vol. 5, 177–279.
- ATALAY, E., E. FROST, A. SORENSEN, C. SULLIVAN, AND W. ZHU (2025): “Scalable Demand and Markups,” *Journal of Political Economy*, Forthcoming.
- ATALAY, E., A. SORENSEN, C. SULLIVAN, AND W. ZHU (2024): “Product Repositioning by Merging Firms,” *Journal of Industrial Economics*, 72, 868–908.
- ATHEY, S., J. LEVIN, AND E. SEIRA (2011): “Comparing Open and Sealed Bid Auctions: Evidence from Timber Auctions,” *Quarterly Journal of Economics*, 126, 207–257.
- BACKUS, M., C. CONLON, AND M. SINKINSON (2021): “Common Ownership and Competition in the Ready-to-Eat Cereal Industry,” Tech. rep., National Bureau of Economic Research.
- BAJARI, P. AND A. HORTACSU (2005): “Are Structural Estimates of Auction Models Reasonable? Evidence from Experimental Data,” *Journal of Political Economy*, 113, 703–741.
- BERRY, S., J. LEVINSOHN, AND A. PAKES (1995): “Automobile Prices in Market Equilibrium,” *Econometrica*, 63, 841–890.
- (2004): “Differentiated products demand systems from a combination of micro and macro data: The new car market,” *Journal of Political Economy*, 112, 68–105.
- BHATTACHARYA, V., G. ILLANES, AND D. STILLERMAN (2025): “Merger Effects and Antitrust Enforcement: Evidence from US Consumer Packaged Goods,” Tech. rep., Working Paper.
- BJÖRNERSTEDT, J. AND F. VERBOVEN (2016): “Does Merger Simulation Work? Evidence from the Swedish Analgesics Market,” *American Economic Journal: Applied Economics*, 8, 125–64.
- BONNET, C. AND P. DUBOIS (2010): “Inference on Vertical Contracts Between Manufacturers and Retailers Allowing for Nonlinear Pricing and Resale Price Maintenance,” *RAND Journal of Economics*, 41, 139–164.
- BRAGUINSKY, S., A. OHYAMA, T. OKAZAKI, AND C. SYVERSON (2015): “Acquisitions, Productivity, and Profitability: Evidence from the Japanese Cotton Spinning Industry,” *American Economic Review*, 105, 2086–2119.
- BRAND, J. (2021): “Differences in Differentiation: Rising Variety and Markups in Retail Food Stores,” Tech. rep., Microsoft Research.
- BRESNAHAN, T. F. (1987): “Competition and Collusion in the American Automobile Industry: The 1955 Price War,” *Journal of Industrial Economics*, 457–482.
- BROT, Z., Z. COOPER, S. V. CRAIG, AND L. KLARNET (2024): “Is There Too Little Antitrust Enforcement in the US Hospital Sector?” *American Economic Review: Insights*, 6, 526–542.
- BUTTERS, R. A., D. W. SACKS, AND B. SEO (2022): “How Do National Firms Respond to Local Cost Shocks?” *American Economic Review*, 112, 1737–1772.

- CHAMBERLAIN, G. (1987): “Asymptotic Efficiency in Estimation with Conditional Moment Restrictions,” *Journal of Econometrics*, 34, 305–334.
- CONLON, C. AND J. GORTMAKER (2020): “Best Practices for Differentiated Products Demand Estimation with pyblp,” *RAND Journal of Economics*, 51, 1108–1161.
- (2025): “Incorporating micro data into differentiated products demand estimation with PyBLP,” *Journal of Econometrics*, 105926.
- CRAWFORD, G. S., R. S. LEE, M. D. WHINSTON, AND A. YURUKOGLU (2018): “The Welfare Effects of Vertical Integration in Multichannel Television Markets,” *Econometrica*, 86, 891–954.
- DEARING, A., L. MAGNOLFI, D. QUINT, C. J. SULLIVAN, AND S. B. WALDFOGEL (2024): “Learning Firm Conduct: Pass-Through as a Foundation for Instrument Relevance,” Tech. rep., National Bureau of Economic Research.
- DELLAVIGNA, S. AND M. GENTZKOW (2019): “Uniform Pricing in US Retail Chains,” *Quarterly Journal of Economics*, 134, 2011–2084.
- DEMIRER, M. AND O. KARADUMAN (2025): “Do Mergers and Acquisitions Improve Efficiency: Evidence from Power Plants,” *Journal of Political Economy*, Forthcoming.
- DÖPPER, H., A. MACKAY, N. H. MILLER, AND J. STIEBALE (2025): “Rising Markups and the Role of Consumer Preferences,” *Journal of Political Economy*, 133, 2462–2505.
- DRANOVE, D. AND C. ODY (2016): “Evolving Measures of Provider Market Power,” *American Journal of Health Economics*, 2, 145–160.
- DUARTE, M., L. MAGNOLFI, M. SØLVSTEN, AND C. SULLIVAN (2024): “Testing Firm Conduct,” *Quantitative Economics*, 15, 571–606.
- DUBOIS, P., R. GRIFFITH, AND M. O’CONNELL (2018): “The Effects of Banning Advertising in Junk Food Markets,” *Review of Economic Studies*, 85, 396–436.
- ESCUADERO, M. (2018): “Rule-of-Thumb Pricing: Retail Cannabis in Washington State,” Tech. rep., Working Paper.
- EUROMONITOR INTERNATIONAL (2022): “Euromonitor Passport,” Various data series accessed by hand from interactive website. Last accessed August 2022.
- FEDERAL RESERVE BANK OF ST. LOUIS (2025): “Federal Reserve Economic Data,” Various series described in README. Last accessed August 2024.
- FOCARELLI, D. AND F. PANETTA (2003): “Are Mergers Beneficial to Consumers? Evidence from the Market for Bank Deposits,” *American Economic Review*, 93, 1152–1172.
- FOURNIER, G. M. AND Y. GAI (2007): “What does Willingness-to-Pay Reveal About Hospital Market Power in Merger Cases?” in *IHEA 2007 6th World Congress*.
- GANDHI, A. AND J.-F. HOUDE (2019): “Measuring Substitution Patterns in Differentiated Products Industries,” *Working Paper*.
- GARMON, C. (2017): “The Accuracy of Hospital Merger Screening Methods,” *RAND Journal of Economics*, 48, 1068–1102.
- GRIECO, P., J. PINKSE, AND M. SLADE (2018): “Brewed in North America: Mergers, Marginal Costs, and Efficiency,” *International Journal of Industrial Organization*, 59, 24–65.
- HAUSMAN, J., G. LEONARD, AND J. D. ZONA (1994): “Competitive Analysis with Differentiated Products,” *Annales d’Economie et de Statistique*, 159–180.
- HENDEL, I. AND A. NEVO (2006): “Measuring the Implications of Sales and Consumer Inventory Behavior,” *Econometrica*, 74, 1637–1673.
- JEZIORSKI, P. (2014): “Estimation of Cost Efficiencies from Mergers: Application to US Radio,” *RAND Journal of Economics*, 45, 816–846.
- KABOSKI, J. P. AND R. M. TOWNSEND (2011): “A Structural Evaluation of a Large-Scale Quasi-Experimental Microfinance Initiative,” *Econometrica*, 79, 1357–1406.

- KEHOE, T. J., P. S. PUJOLAS, AND J. ROSSBACH (2017): “Quantitative Trade Models: Developments and Challenges,” *Annual Review of Economics*, 9, 295–325.
- KIM, E. H. AND V. SINGAL (1993): “Mergers and Market Power: Evidence from the Airline Industry,” *American Economic Review*, 549–569.
- MACKAY, A. AND N. H. MILLER (2025): “Estimating Models of Supply and Demand: Instruments and Covariance Restrictions,” *American Economic Journal: Microeconomics*, 17, 238–281.
- MAGNOLFI, L., J. MCCLURE, AND A. SORENSEN (2025): “Triplet Embeddings for Demand Estimation,” *American Economic Journal: Microeconomics*, 17, 282–307.
- MAY, S. AND M. NOETHER (2014): “Unresolved Questions Relating to Market Definition in Hospital Mergers,” *Antitrust Bulletin*, 59, 479–503.
- MILLER, N. H. AND M. C. WEINBERG (2017): “Understanding the Price Effects of the MillerCoors Joint Venture,” *Econometrica*, 85, 1763–1791.
- MINCER, J. A. AND V. ZARNOWITZ (1969): “The Evaluation of Economic Forecasts,” in *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER, 3–46.
- MORROW, W. R. AND S. J. SKERLOS (2011): “Fixed-Point Approaches to Computing Bertrand-Nash Equilibrium Prices Under Mixed-Logit Demand,” *Operations Research*, 59, 328–345.
- MOTTA, M. AND E. TARANTINO (2021): “The Effect of Horizontal Mergers, When Firms Compete in Prices and Investments,” *International Journal of Industrial Organization*, 78, 102774.
- NAKAMURA, E. AND D. ZEROM (2010): “Accounting for Incomplete Pass-Through,” *Review of Economic Studies*, 77, 1192–1230.
- NEVO, A. (2000): “Mergers with Differentiated Products: The Case of the Ready-to-Eat Cereal Industry,” *RAND Journal of Economics*, 395–421.
- (2001): “Measuring Market Power in the Ready-to-Eat Cereal Industry,” *Econometrica*, 69, 307–342.
- NEVO, A. AND M. D. WHINSTON (2010): “Taking the Dogma out of Econometrics: Structural Modeling and Credible Inference,” *Journal of Economic Perspectives*, 24, 69–82.
- NIELSEN IQ (2025): “Consumer Panel and Retail Scanner Data,” Multiple electronic files obtained from the Kilts Center at the University of Chicago.
- PANHANS, M. T. AND C. TARAGIN (2023): “Consequences of Model Choice in Predicting Horizontal Merger Effects,” *International Journal of Industrial Organization*, 89, 102986.
- PATHAK, P. A. AND P. SHI (2021): “How Well Do Structural Demand Models Work? Counterfactual Predictions in School Choice,” *Journal of Econometrics*, 222, 161–195.
- PETERS, C. (2006): “Evaluating the Performance of Merger Simulation: Evidence from the US Airline Industry,” *Journal of Law and Economics*, 49, 627–649.
- PORTER, R. H. (1983): “A Study of Cartel Stability: The Joint Executive Committee, 1880-1886,” *Bell Journal of Economics*, 301–314.
- REYNAERT, M. AND F. VERBOVEN (2014): “Improving the Performance of Random Coefficients Demand Models: The Role of Optimal Instruments,” *Journal of Econometrics*, 179, 83–98.
- SALANIÉ, B. AND F. A. WOLAK (2022): “Fast, detail-free, and approximately correct: Estimating mixed demand systems,” Tech. rep., Working paper.
- SCHMITT, M. (2017): “Do Hospital Mergers Reduce Costs?” *Journal of Health Economics*, 52, 74–94.
- SHEEN, A. (2014): “The Real Product Market Impact of Mergers,” *Journal of Finance*, 69, 2651–2688.
- SULLIVAN, C. (2020): “The Ice Cream Split: Empirically Distinguishing Price and Product Space Collusion,” Tech. rep., University of Alberta.
- THOMSON REUTERS (2021): “SDC Platinum,” Data extracted from electronic database. Last accessed January 2021.
- WEINBERG, M. C. (2011): “More Evidence on the Performance of Merger Simulations,” *American Economic Review Papers and Proceedings*, 101, 51–55.

- WEINBERG, M. C. AND D. HOSKEN (2013): “Evidence on the Accuracy of Merger Simulations,” *Review of Economics and Statistics*, 95, 1584–1600.
- WERDEN, G. J. AND L. M. FROEB (1994): “The Effects of Mergers in Differentiated Products Industries: Logit Demand and Merger Policy,” *Journal of Law, Economics, and Organization*, 10, 407–426.
- WERDEN, G. J. AND L. M. FROEB (2006): “Unilateral Competitive Effects of Horizontal Mergers,” in *Handbook of Antitrust Economics*, MIT Press.
- WHINSTON, M. D. (2007): “Antitrust Policy Toward Horizontal Mergers,” in *Handbook of Industrial Organization*, ed. by M. Armstrong and R. Porter, Elsevier, vol. 3, 2369–2440.

ONLINE APPENDIX

A. Additional Tables and Figures

This appendix contains additional figures and tables mentioned in the text.

1. Figure A.1 shows diversion ratios between the target and acquirer, for the estimated model (red) and a logit model (blue).
2. Table A.1 compares estimated elasticities to those from other papers. Keep in mind that the definition of a product and a market are different for different papers (as are the time periods and geographies under study), so these comparisons should be made with caution.

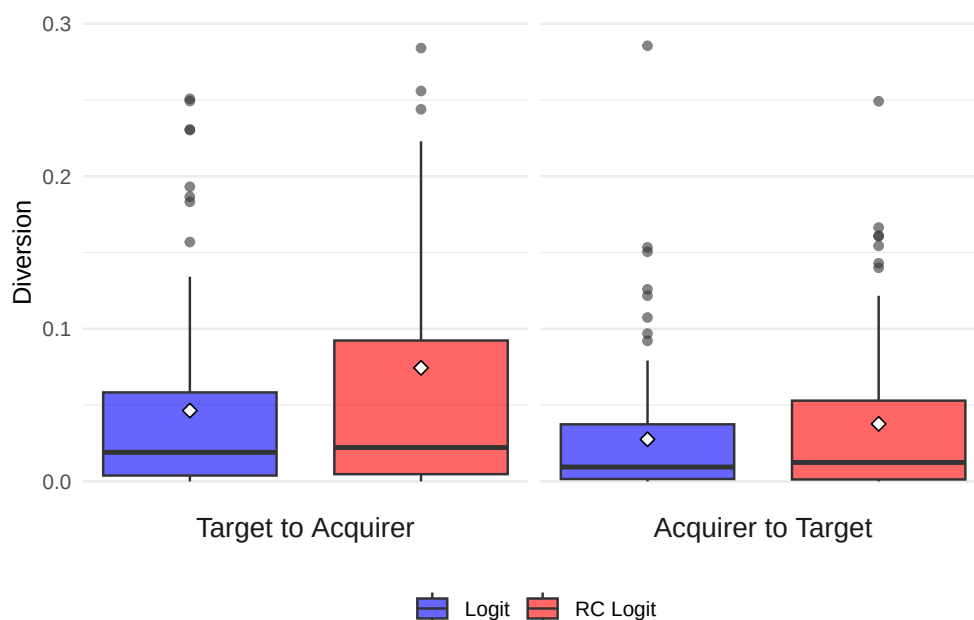


Figure A.1: Comparison of diversion ratios between merging parties under logit and random coefficients logit. The white diamonds are the mean across mergers, while the lower, middle, and upper lines of the boxes are the 25th, 50th, and 75th percentiles.

Product Market	Estimates from Literature	Our Estimates
Ready-to-eat cereal	-2.3 to -4.3 (Nevo, 2001, Table VIII) -2.4 to -2.7 (Backus et al., 2021, Table 4) -3.9 to -4.2 (Atalay et al., 2025)	-2.3 (RTE cereal) -4.8 (granola)
Laundry detergent	-2.4 to -4.3 (Hendel and Nevo, 2006, Table VII) -5.3 to -10.0 (Atalay et al., 2025)	-2.7 (detergent boosters)
Beer	-4.3 to -6.3 (Miller and Weinberg, 2017, Section 4.3) -3.4 (Asker, 2016, Table III)	-3.4 to -3.9
Coffee	-3.5 (Nakamura and Zerom, 2010, Table 5) -4.6 to -5.6 (Atalay et al., 2025)	-1.6
Salted Snacks	-1.5 to -3.0 (chips) (Dubois et al., 2018, Table D.5) -4.5 to -8.4 (chips) (Atalay et al., 2025) -3.0 to -5.3 (pretzels) (Atalay et al., 2025)	-12.8 (chips) -3.1 (popcorn) -2.5 to -7.6 (pretzels)
Wine	-3.6 to -5.3 (Atalay et al., 2025)	-2.4 to -10.0
Bottled water	-5.0 to -13.0 (Bonnet and Dubois, 2010, Table 6) -2.3 to -6.8 (Atalay et al., 2025)	-3.5
Frozen pizza	-3.5 to -4.3 (Atalay et al., 2025)	-3.0
Frozen breakfasts	-2.6 to -3.7 (Atalay et al., 2025)	-3.9
Sliced lunchmeat	-1.5 to -3.9 (Atalay et al., 2025)	-3.6
Bakery products	-2.1 to -6.5 (Atalay et al., 2025)	-1.4 to -3.3 (-4.4 for fresh bread)
Cookies	-3.8 to -4.8 (Atalay et al., 2025)	-1.2
Breakfast sausages	-4.6 to -6.5 (Atalay et al., 2025)	-7.6

Table A.1: Comparisons of own-price elasticities from our procedure to those of similar markets in the literature. Ranges for our estimates indicate different mergers (with non-overlapping time periods) and ranges for published papers indicate different specifications, different time periods, or different products (if an average elasticity is not provided).

3. Tables A.2 and A.3 show regressions of prediction error on measures of market structure. Figure A.2 shows boxplots of prediction error over time. Figure A.3 overlays prediction errors for placebo mergers: we repeat this simulation assuming a “placebo merger” happened one year prior to the actual merger and predict prices for that year.
4. Figure A.4 shows accuracy of the model in predicting price changes above 0%, 2.5%, or 5%, as a function of the threshold used for the prediction.
5. Table A.4 show a regression of the internalization parameter τ on measures of market structure and market size.
6. Figure A.5 shows the distribution of synergies assuming partial internalization.

	Merging			Overall			Non-Merging		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-0.02*	-0.03***		0.01	-0.00		0.02	0.00	
	(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)	
HHI	-0.03	0.02	-0.01	-0.13***	-0.04	0.00	-0.15***	-0.06*	0.01
	(0.04)	(0.03)	(0.02)	(0.04)	(0.03)	(0.02)	(0.04)	(0.03)	(0.02)
DHHI	0.10	-0.13	-0.47***	0.17	-0.08	-0.40***	0.41***	0.13**	-0.02
	(0.21)	(0.11)	(0.09)	(0.13)	(0.10)	(0.08)	(0.14)	(0.06)	(0.05)
Level	Merger	DMA	DMA	Merger	DMA	DMA	Merger	DMA	DMA
Merger FE	No	No	Yes	No	No	Yes	No	No	Yes

Table A.2: Regression of prediction error on market characteristics, at different levels of observation. Regressions drop any observation where the absolute log prediction error exceeds 0.25.

	Merging			Overall			Non-Merging		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-1.65	-2.42***		-1.55	-1.67***		-1.82	-1.87***	
	(1.19)	(0.27)		(1.27)	(0.21)		(1.45)	(0.23)	
HHI $\in$ [1500, 2500]	-0.50	0.25	-0.30	1.00	1.09***	0.10	1.14	1.34***	0.27
	(1.41)	(0.30)	(0.46)	(1.38)	(0.23)	(0.27)	(1.54)	(0.26)	(0.30)
HHI > 2500	-1.92	0.02	-0.74	-1.93	-0.68***	-0.49	-1.43	-0.58**	-0.20
	(1.39)	(0.31)	(0.59)	(1.46)	(0.25)	(0.45)	(1.63)	(0.27)	(0.42)
DHHI > 100	-0.60	-0.67***	-1.98***	0.07	-0.09	-1.79***	0.92	0.90***	-0.60*
	(1.39)	(0.23)	(0.54)	(1.19)	(0.20)	(0.42)	(1.27)	(0.21)	(0.31)
Level	Merger	DMA	DMA	Merger	DMA	DMA	Merger	DMA	DMA
Merger FE	No	No	Yes	No	No	Yes	No	No	Yes

Table A.3: Regression of prediction error on market characteristics, at different levels of observation. Regressions drop any observation where the absolute log prediction error exceeds 0.25. Coefficients are multiplied by 100.

	(1)	(2)	(3)	(4)	(5)
Constant	0.56***	0.54***	0.52***	1.03***	0.74*
	(0.19)	(0.06)	(0.06)	(0.37)	(0.42)
HHI > 1500	-0.13				-0.10
	(0.19)				(0.19)
DHHI > 100		-0.33***			-0.21*
		(0.09)			(0.12)
Merging Share > 0.30			-0.38***		-0.23**
			(0.08)		(0.11)
Log(Merging Revenue)				-0.04	-0.01
				(0.03)	(0.03)

Table A.4: Regression of the internalization coefficient on measures of market structure

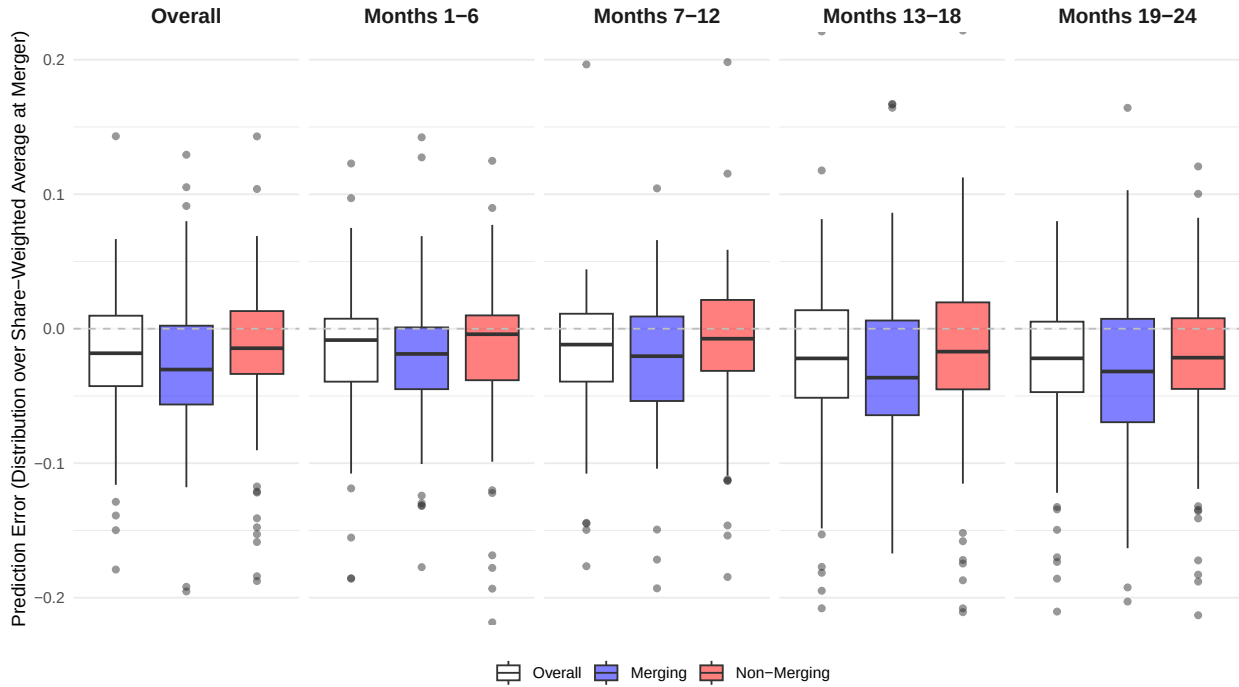


Figure A.2: Distribution of prediction error at the merger level, over time

B. List of All Mergers

Table B.1 summarizes all of the mergers included in our sample. For each merger, we list the NielsenIQ department and product group of the relevant product market, and the NielsenIQ product modules that comprise the relevant product market, along with criteria used for dividing the modules into separate markets (if any). We then define all characteristics that we extract from the scanner data, the share of variance of those characteristics (along with a private-label indicator and the package size, which are included in all mergers) that are explained by the first two principal components, the set of candidate cost shifter instruments, and whether the merger is included in our final estimation sample. For mergers that are included, the table also lists the set of instruments (out of Gandhi-Houde, cost shifters, and Hausman) and specification of demographic interactions (out of Price, 1+Price, and 1+PC+Price) used in the demand estimates included in our sample.

The “cost shifters” column can be read as follows. Each instrument is separated by a semicolon. Generally, these share a name with the FRED series from which they are taken. Sometimes, the instruments are multiplied by a boolean expression of a characteristic, in which case the instrument is interacted with an indicator for whether the boolean is true. If this column is empty then there are no cost shifter instruments defined for this merger and we rely on the other two types of instruments for estimation (differentiation IVs and Hausman instruments).

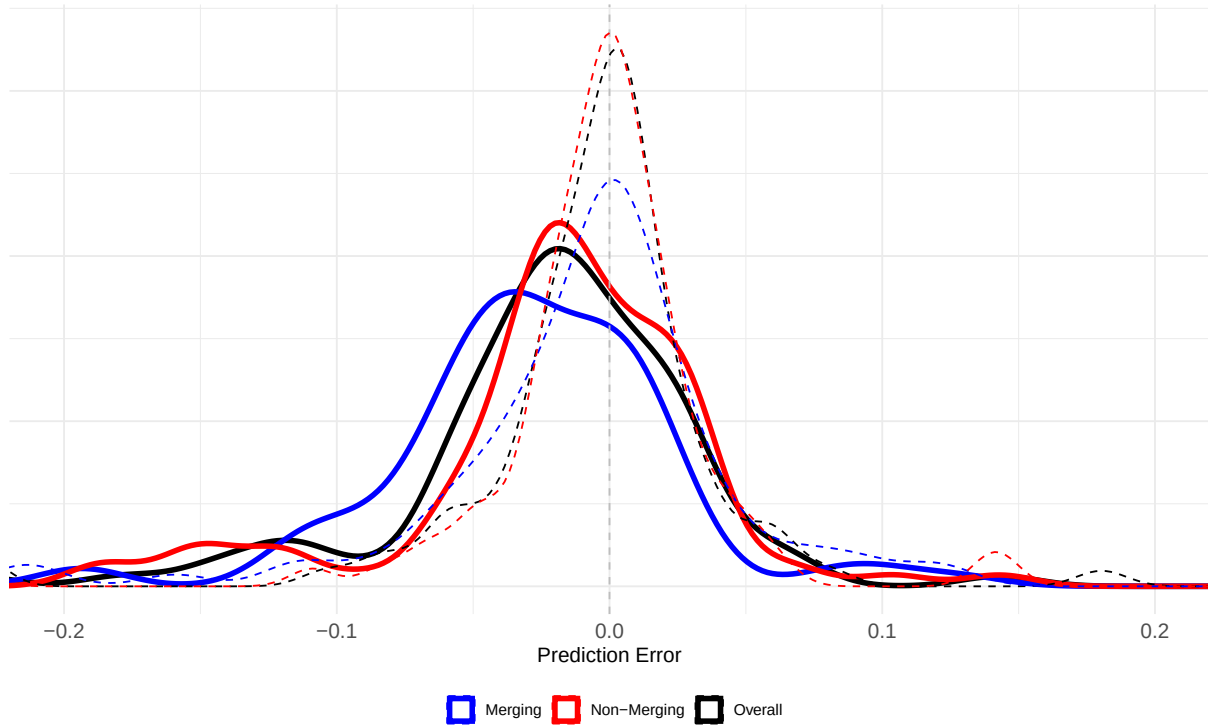


Figure A.3: Distribution of prediction error at the merger level, overlaid with distributions from placebo mergers (dashed)

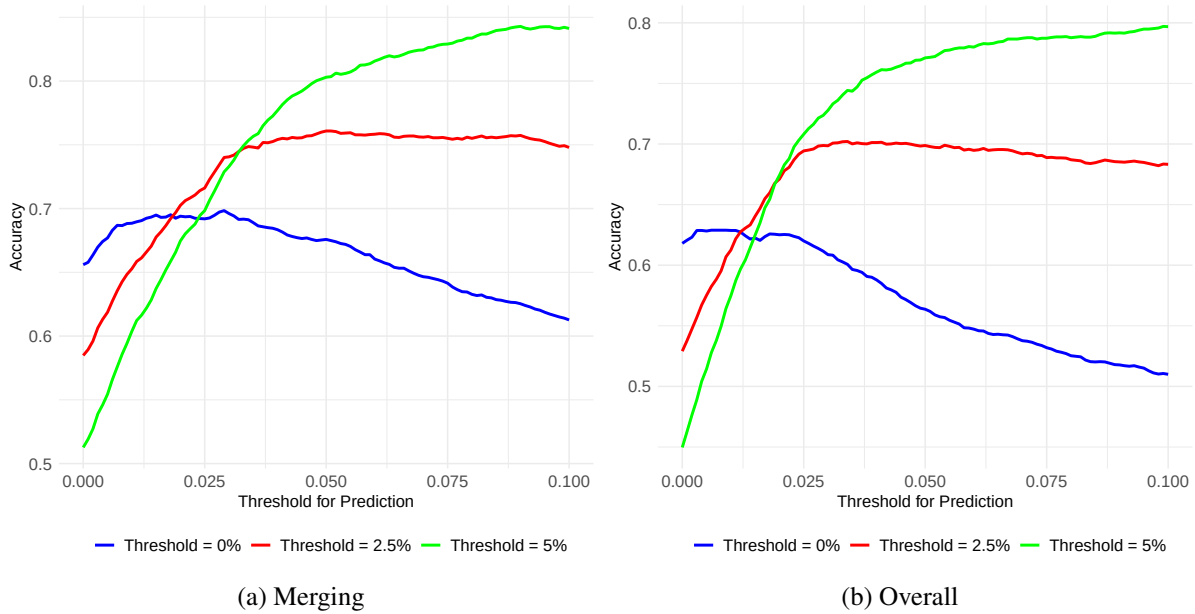


Figure A.4: Accuracy for different thresholds. Each line considered a prediction of the event that the realized price change is larger than 0%, 2.5%, or 5%. The line plots accuracy of predicting this event using the event that the predicted price change is larger than x (on the horizontal axis).

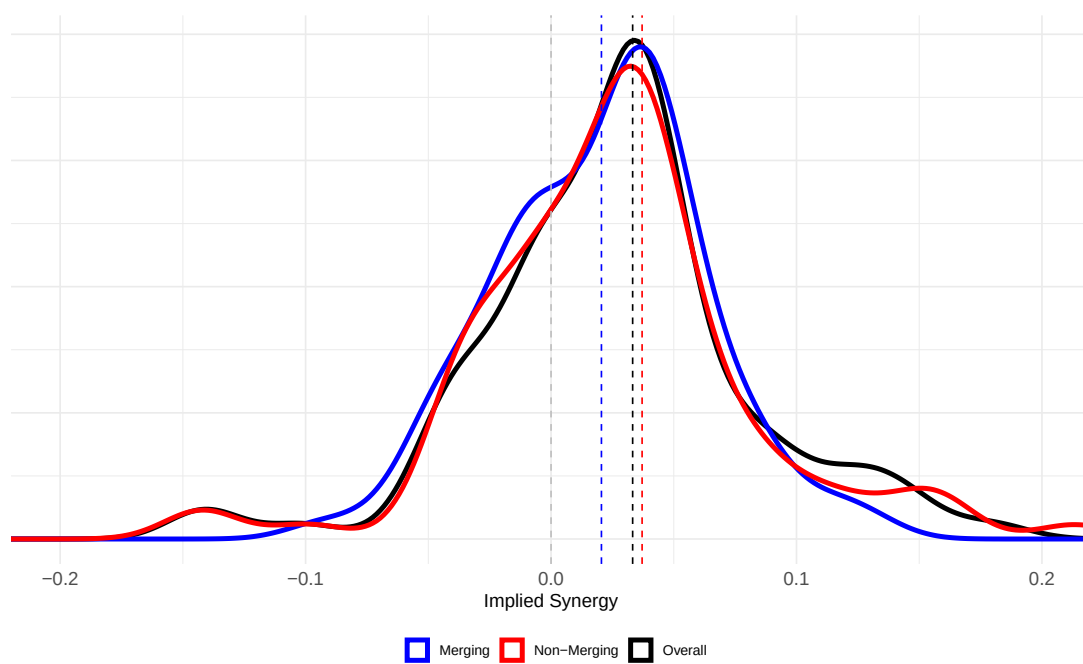


Figure A.5: Distribution of implied synergies at the merger level, assuming partial internalization

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
1	Alcoholic beverages	Beer	Ale; Beer; Light Beer (Low Calorie/Alcohol); Stout And Porter	Container; ice; imported; module	41%	Aluminum x container == "can"; barley; barley x module != "light"; dollar strength x imported == 1; glass x container == "bottle"; wheat	Yes	GH	1+PC+Price
2	Alcoholic beverages	Beer	Ale; Beer; Light Beer (Low Calorie/Alcohol); Stout And Porter	Container; ice; imported; module	50%	Aluminum x container == "can"; barley; barley x module != "light"; dollar strength x imported == 1; glass x container == "bottle"; wheat	Yes	GH+CS	1+PC+Price
3	Alcoholic beverages	Liquor	Alcoholic Cocktails	Flavor; liquor; party pack; quality	28%	Barley x liquor in ["whiskey", "gin"]; corn x liquor == "whiskey"; dollar strength; grapes x liquor == "other"; russet potatoes x liquor == "vodka"; sugar x liquor in ["tequila", "rum"]; wheat x liquor in ["gin", "vodka"]; wine grapes x liquor == "other"	Yes	GH+CS	1+PC+Price
4	Alcoholic beverages	Liquor	Bourbon-Blended; Bourbon-Straight/Bonded; Canadian Whiskey; Irish Whiskey; Remaining Whiskey; Scotch	Imported; liquor type; proof; quality	49%	Barley; corn; dollar strength; dollar strength x imported == 1	Yes	GH+CS	1+PC+Price
5	Alcoholic beverages	Liquor	Bourbon-Blended; Bourbon-Straight/Bonded; Canadian Whiskey; Irish Whiskey; Remaining Whiskey; Scotch	Imported; liquor type; proof; quality	49%	Barley; corn; dollar strength; dollar strength x imported == 1	Yes	GH	1+PC+Price
6	Alcoholic beverages	Liquor	Cordials & Proprietary Liqueurs	Alcohol content; flavor; imported; liquor	28%	Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac"	Yes	GH	1+PC+Price
7	Alcoholic beverages	Liquor	Cordials & Proprietary Liqueurs	Alcohol content; flavor; imported; liquor	28%	Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac"	Yes	GH	1+PC+Price
8	Alcoholic beverages	Liquor	Cordials & Proprietary Liqueurs	Alcohol content; flavor; imported; liquor	27%	Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac"	Yes	GH	1+PC+Price
9	Alcoholic beverages	Liquor	Gin	Flavored; gin type; imported; quality	46%	Barley; dollar strength; dollar strength x imported == 1; wheat	Yes	GH	1+PC+Price
10	Alcoholic beverages	Liquor	Rum	Flavored; imported; proof; quality; type of rum	42%	Dollar strength; dollar strength x imported == 1; sugar	Yes	GH	1+PC+Price
11	Alcoholic beverages	Liquor	Tequila	Flavored; proof; quality; type of tequila	39%	Dollar strength; sugar	Yes	GH+CS	1+PC+Price
12	Alcoholic beverages	Liquor	Vodka	Flavored; imported; proof; quality	53%	Dollar strength; dollar strength x imported == 1; russet potatoes; wheat	Yes	GH+CS	1+PC+Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
13	Alcoholic beverages	Liquor	Vodka	Flavored; imported; proof; quality	53%	Dollar strength; dollar strength x imported == 1; russet potatoes; wheat	Yes	GH+CS	1+PC+Price
14	Alcoholic beverages	Wine	Wine-Domestic Dry Table; Wine-Imported Dry Table; is red = 0	Grape type; imported; is red; quality	37%	Dollar strength x imported == 1; wine grapes	Yes	GH+CS	1+PC+Price
15	Alcoholic beverages	Wine	Wine-Domestic Dry Table; Wine-Imported Dry Table; is red = 1	Grape type; imported; is red; quality	32%	Dollar strength x imported == 1; wine grapes	Yes	GH+H	1+PC+Price
16	Dry grocery	Baby Food	Baby Milk And Milk Flavoring; infant and toddler = 0	Form; infant and toddler; organic	47%	Corn sweeteners; starch vegetable fats oils; vitamin nutrient hematinic human	Yes	GH	Price
17	Dry grocery	Baby Food	Baby Milk And Milk Flavoring; infant and toddler = 1	Form; infant and toddler; organic	77%	Corn sweeteners; starch vegetable fats oils; vitamin nutrient hematinic human	Yes	GH+H	1+Price
18	Dry grocery	Bread And Baked Goods	Bakery - Bread - Fresh	Bread type; bread usage; is herb garlic; is potato; is sourdough; is sweet; is wheat; low cholesterol; low fat; regular health	33%	Russet potatoes x is potato == 1; wheat	Yes	GH+CS	1+Price
19	Dry grocery	Bread And Baked Goods	Bakery - Bread - Fresh	Bread type; bread usage; is herb garlic; is potato; is sourdough; is sweet; is wheat; low cholesterol; low fat; regular health	32%	Russet potatoes x is potato == 1; wheat	Yes	GH	Price
20	Dry grocery	Bread And Baked Goods	Bakery-Bagels-Fresh	Bagel size; is low fat; primary flavor; quality tier; secondary flavor	8%	Wheat; wheat flour	Yes	GH+H	1+PC+Price
21	Dry grocery	Bread And Baked Goods	Bakery-Bagels-Fresh	Bagel size; has no salt; is low fat; is sliced; is wheat; primary flavor; quality tier; secondary flavor	9%	Sugar x primary flavor == "sweet"; wheat; wheat flour	Yes	GH	Price
22	Dry grocery	Bread And Baked Goods	Bakery-Breakfast Cakes/Sweet Rolls-Fresh	Has nuts; is filled; primary flavor; primary type; quality tier; secondary flavor; secondary type	6%	Cocoa beans x primary flavor == "chocolate"; corn sweeteners; sugar; wheat; wheat flour	Yes	GH	1+PC+Price
23	Dry grocery	Bread And Baked Goods	Bakery-Buns-Fresh	Ingredient; is low cholesterol; is low fat; is potato; is specialty; is wheat; primary flavor; primary type; secondary flavor; secondary type	10%	Russet potatoes x is potato == 1; wheat	Yes	GH+CS	1+PC+Price
24	Dry grocery	Bread And Baked Goods	Bakery-Buns-Fresh	Ingredient; is low cholesterol; is low fat; is potato; is specialty; is wheat; primary flavor; primary type; secondary flavor; secondary type	9%	Russet potatoes x is potato == 1; wheat	Yes	GH+H	1+PC+Price
25	Dry grocery	Bread And Baked Goods	Bakery-Doughnuts-Fresh	Donut style; has chocolate; is frosted; is glazed; primary flavor; secondary flavor	11%	Cocoa beans x has chocolate == 1; corn sweeteners; sugar; vegetable oil; wheat	Yes	GH	1+PC+Price
26	Dry grocery	Bread And Baked Goods	Bakery-Doughnuts-Fresh	Donut style; has chocolate; is frosted; is glazed; primary flavor; secondary flavor	10%	Cocoa beans x has chocolate == 1; corn sweeteners; sugar; vegetable oil; wheat	Yes	GH	1+PC+Price
27	Dry grocery	Bread And Baked Goods	Bakery-Muffins-Fresh	Is english muffin; is mini; is wheat; low cholesterol; low fat; primary flavor; quality tier; secondary flavor	8%	Cocoa beans x primary flavor == "chocolate dessert"; other grains x primary flavor == "breakfast grains"; sugar x is english muffin == 0; wheat; wheat flour	Yes	GH+H	1+PC+Price
28	Dry grocery	Bread And Baked Goods	Bakery-Pies-Fresh	Is tiny; primary flavor; secondary flavor	21%		Yes	GH	1+Price
29	Dry grocery	Bread And Baked Goods	Bakery-Rolls-Fresh	Bread region; is low cholesterol; is low fat; is potato; is sourdough; is wheat; primary flavor; primary type; secondary flavor; secondary type	3%	Russet potatoes x is potato == 1; wheat	Yes	GH+H	1+PC+Price
30	Dry grocery	Bread And Baked Goods	Bakery-Rolls-Fresh	Bread region; is low cholesterol; is low fat; is potato; is sourdough; is wheat; primary flavor; primary type; secondary flavor; secondary type	4%	Russet potatoes x is potato == 1; wheat	Yes	GH+H	1+PC+Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
31	Dry grocery	Candy	Candy-Chocolate; Candy-Chocolate-Miniatures; Candy-Chocolate-Special	Chocolate shape; chocolate type; flavor; high quality; holiday; module info	21%	Almonds x flavor == "almond"; cocoa beans; cocoa beans x chocolate type != "white"; corn sweeteners x flavor == "caramel"; peanuts x flavor == "peanut butter"; sugar; sugar x chocolate type != "bittersweet"	No		
32	Dry grocery	Candy	Candy-Dietetic - Chocolate; Candy-Dietetic - Non-Chocolate	Primary type	45%	Corn sweeteners; sugar	No		
33	Dry grocery	Candy	Candy-Hard Rolled; Candy-Lollipops; Candy-Non-Chocolate; Candy-Non-Chocolate-Miniatures	Candy type	52%	Corn sweeteners; sugar	No		
34	Dry grocery	Candy	Candy-Hard Rolled; Candy-Lollipops; Candy-Non-Chocolate; Candy-Non-Chocolate-Miniatures	Candy type; flavor; holiday	28%	Corn sweeteners; sugar	No		
35	Dry grocery	Cereal	Cereal - Granola & Natural Types	Has chocolate; high fiber; high protein; low sugar; nut type; primary flavor; secondary flavor	11%	Almonds x nut type == "almond"; other grains; peanuts x nut type == "peanut"; pecans x nut type == "pecan"; sugar x low sugar == 0	Yes	GH	1+PC+Price
36	Dry grocery	Cereal	Cereal - Ready To Eat	Additions; audience; high fiber; ingredient; is sweet; low cholesterol; low sugar	38%	Corn; corn sweeteners; corn sweeteners x is sweet; corn x ingredient == "corn"; oats; oats x ingredient == "oats"; rough rice; rough rice x ingredient == "rice"; sugar; sugar x is sweet; wheat; wheat x ingredient in ["wheat", "multigrain/other"]	Yes	GH	1+Price
37	Dry grocery	Coffee	Coffee - Soluble; Coffee - Soluble Flavored	Decaf; drying method; flavor; is flavored; milk beverage; roast	37%	Cocoa beans x flavor == "chocolate"; coffee beans; dairy x milk beverage == 1	No		
38	Dry grocery	Coffee	Ground And Whole Bean Coffee; cup = 0	Cup; decaf; flavored	59%	Coffee beans	Yes	GH+H	1+Price
39	Dry grocery	Coffee	Ground And Whole Bean Coffee; cup = 1	Cup; decaf; flavored	58%	Coffee beans	Yes	GH+H	Price
40	Dry grocery	Condiments, Gravies, And Sauces	Fish & Seafood & Cocktail Sauce	Sauce type	46%	Mayonnaise and dressing x sauce type == "tartar"; pickles and horseradish x sauce type == "tartar"; shrimp; tomatos x sauce type == "cocktail"; unprocessed finfish	Yes	GH	1+PC+Price
41	Dry grocery	Condiments, Gravies, And Sauces	Meat Sauce; Worcestershire Sauce	Flavor; sauce type	41%	Salt pepper spices; tomatos x sauce type == "steak"; vinegar	No		
42	Dry grocery	Condiments, Gravies, And Sauces	Mustard	Glass container; mustard type	31%	Glass x glass container == 1; mustard seed futures x mustard type != "horseradish wasabi"; salt pepper spices; vinegar	No		
43	Dry grocery	Condiments, Gravies, And Sauces	Sauce & Seasoning Mix-Remaining	Cuisine; flavor; preparation	31%	Dry onions; salt pepper spices; spices; tomatos; vinegar	Yes	GH	1+PC+Price
44	Dry grocery	Condiments, Gravies, And Sauces	Sauce & Seasoning Mix-Remaining	Bbq; beef; burger; curry; pork; sausage; turkey; wing	32%	Dry onions; salt pepper spices; spices; tomatos; vinegar	Yes	GH	Price
45	Dry grocery	Condiments, Gravies, And Sauces	Sauce & Seasoning Mix-Remaining Mexican; Sauce Mix - Taco	Hot; other flavor; type of sauce	28%	Dry onions; salt pepper spices; spices; tomatos x type of sauce == "salsa"; vinegar	Yes	GH+CS	1+PC+Price
46	Dry grocery	Condiments, Gravies, And Sauces	Sauce Mix - Spaghetti	Mushroom; pesto or alfredo	61%	Salt pepper spices; spices; tomatos	Yes	GH	1+PC+Price
47	Dry grocery	Condiments, Gravies, And Sauces	Seasoning Mix - Chili	Hot; regular; texas	65%	Dry onions; salt pepper spices; spices; tomatos	Yes	GH+CS	1+PC+Price
48	Dry grocery	Condiments, Gravies, And Sauces	Seasoning Mix - Chili	Hot; regular; texas	63%	Dry onions; salt pepper spices; spices; tomatos; vinegar	Yes	GH	1+PC+Price
49	Dry grocery	Condiments, Gravies, And Sauces	Seasoning Mix - Sloppy Joe	No msg	87%	Dry onions; salt pepper spices; spices; tomatos	Yes	GH+H	1+PC+Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
50	Dry grocery	Cookies	Cookies	Bite size; coated; cookie type; flavor; holiday; low fat	22%		Yes	GH+H	1+PC+Price
51	Dry grocery	Gum	Gum-Bubble; Gum-Bubble-Sugarfree; Gum-Chewing; Gum-Chewing-Sugarfree	Bubblegum; flavor; presentation; sugarfree	38%	Corn sweeteners x sugarfree == 0; resin and synthetic rubber; sugar x sugarfree == 0	Yes	GH+CS	Price
52	Dry grocery	Gum	Gum-Bubble; Gum-Bubble-Sugarfree; Gum-Chewing; Gum-Chewing-Sugarfree	Flavor; sugarfree	42%	Corn sweeteners x sugarfree == 0; resin and synthetic rubber; sugar x sugarfree == 0	No		
53	Dry grocery	Jams, Jellies, Spreads	Garlic Spreads	Herbs	93%	Soybean oil; vegetable oil	No		
54	Dry grocery	Pet Food	Cat Food - Dry Type	Bulk; flavor; kitten; quality	47%	Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat	Yes	GH	1+PC+Price
55	Dry grocery	Pet Food	Dog & Cat Treats; animal = big dog	Animal; flavor	52%	Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"	Yes	GH+CS	1+PC+Price
56	Dry grocery	Pet Food	Dog & Cat Treats; animal = cat	Animal; flavor	53%	Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"	Yes	GH+CS	1+PC+Price
57	Dry grocery	Pet Food	Dog & Cat Treats; animal = other	Animal; flavor	60%	Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"	Yes	GH	1+PC+Price
58	Dry grocery	Pet Food	Dog & Cat Treats; animal = small dog	Animal; flavor	45%	Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"	Yes	GH+CS	1+PC+Price
59	Dry grocery	Pet Food	Dog Food - Dry Type	Bulk; flavor; puppy; quality	36%	Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat	No		
60	Dry grocery	Pet Food	Dog Food - Moist Type; Dog Food - Wet Type; puppy = 0	Flavor; puppy	46%	Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat	No		
61	Dry grocery	Pet Food	Dog Food - Moist Type; Dog Food - Wet Type; puppy = 1	Flavor; puppy	75%	Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat	No		
62	Dry grocery	Pickles, Olives, And Relish	Relishes	Condiment type; dill; fruity; hot; sweet	52%	Cucumbers x condiment type == "pickle relish"; sugar x sweet == 1; vinegar	Yes	GH	1+PC+Price
63	Dry grocery	Pickles, Olives, And Relish	Relishes	Condiment type; dill; fruity; hot; sweet	49%	Cucumbers x condiment type == "pickle relish"; sugar x sweet == 1; vinegar	Yes	GH	1+PC+Price
64	Dry grocery	Prepared Food-Ready-To-Serve	Chicken - Shelf Stable	Chicken type; has broth; white only	40%	Aluminum; poultry processing; slaughter poultry; wheat x chicken type in ["noodle", "pot pie", "dumplings"]	Yes	GH	1+PC+Price
65	Dry grocery	Prepared Food-Ready-To-Serve	Chili-Shelf Stable	Con carne; has beans; hot; meat	32%	Aluminum; beans city average x has beans == 1; poultry processing x meat in ["chicken", "turkey"]; slaughter cattle x meat == "regular"; slaughter poultry x meat in ["chicken", "turkey"]	Yes	GH	1+PC+Price
66	Dry grocery	Prepared Food-Ready-To-Serve	Stew - Beef - Shelf Stable; Stew - Chicken - Shelf Stable; Stew - Remaining - Shelf Stable	Brunswick; contains beef; contains chicken; microwave; regular	57%	Beans city average x brunswick == 1; poultry processing x contains chicken == 1; sauces; slaughter cattle x contains beef == 1; slaughter poultry x contains chicken == 1; tomatoes x brunswick == 1	Yes	GH+H	1+PC+Price
67	Dry grocery	Shortening, Oil	Cooking Sprays	Butter; oil type; pump	44%	Dairy x butter == 1; olive oil x oil type == "olive"; soybean oil x oil type == "soybean"; vegetable oil x oil type == "vegetable"	Yes	GH	Price
68	Dry grocery	Snacks	Dip - Mixes	Type of mix	34%	Dry onions; salt pepper spices	Yes	GH	1+PC+Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
69	Dry grocery	Snacks	Popcorn - Popped; Snacks - Caramel Corn	Flavor; is caramel corn; low fat; low salt	34%	Cheese x flavor == "cheese"; corn; corn sweeteners x is caramel corn == 1; sugar x flavor == "sweet"	No		
70	Dry grocery	Snacks	Popcorn - Popped; Snacks - Caramel Corn	Flavor; is caramel corn; low fat; low salt	31%	Cheese x flavor == "cheese"; corn; corn sweeteners x is caramel corn == 1; sugar x flavor == "sweet"	Yes	GH	1+PC+Price
71	Dry grocery	Snacks	Snacks - Potato Chips; Snacks - Potato Sticks	Low fat; low salt	57%	Russet potatoes; salt pepper spices; vegetable oil; vegetable oil x low fat == 0	Yes	GH	1+PC+Price
72	Dry grocery	Snacks	Snacks - Pretzel	Flavor; is coated; is filled; low fat; low salt; pretzel size	32%	Peanuts x flavor == "nutty"; sugar x flavor == "sweet"; wheat; wheat flour	Yes	GH	1+PC+Price
73	Dry grocery	Snacks	Snacks - Pretzel	Flavor; is coated; is filled; low fat; low salt; pretzel size	32%	Peanuts x flavor == "nutty"; sugar x flavor == "sweet"; wheat; wheat flour	Yes	GH	1+Price
74	Dry grocery	Snacks	Snacks - Remaining	Flavor; ingredient; low fat; low salt; snack preparation; snack type	19%	Beans city average x ingredient == "legume"; corn x ingredient == "corn"; rough rice x ingredient == "rice"; russet potatoes x ingredient == "potato"; wheat flour x ingredient == "wheat grain"; wheat x ingredient == "wheat grain"	No		
75	Dry grocery	Snacks	Snacks - Remaining	Flavor; ingredient; low fat; low salt; snack preparation; snack type	18%		No		
76	Dry grocery	Soft Drinks-Non-Carbonated	Water-Bottled	Flavored; single bottle; type of water	46%	Diesel; plastic; vitamins x type of water == "enhanced"	Yes	GH	1+PC+Price
77	Dry grocery	Spices, Seasoning, Extracts	Meat Marinades & Tenderizers	Asian; citrus or mojo; herb and garlic; mesquite bbq; steakhouse; tenderizer; teriyaki	32%	Lemons x citrus or mojo == 1; salt pepper spices; spices	Yes	GH	1+PC+Price
78	Dry grocery	Spices, Seasoning, Extracts	Pepper	Black; garlic; ground; lemon; red; whole	42%	Lemons x lemon == 1; salt pepper spices; spices	Yes	GH	1+PC+Price
79	Dry grocery	Spices, Seasoning, Extracts	Salt - Cooking/Edible/Seasoned	Garlic; kosher; seasoned	63%		Yes	GH	1+Price
80	Dry grocery	Spices, Seasoning, Extracts	Seasoning-Dry; spice = All purpose seasoning or blend	Spice	100%		Yes	GH	Price
81	Dry grocery	Spices, Seasoning, Extracts	Seasoning-Dry; spice = Chili powder	Spice	100%		Yes	GH+H	1+PC+Price
82	Dry grocery	Spices, Seasoning, Extracts	Seasoning-Dry; spice = Cinnamon powder	Spice	100%		Yes	GH+H	1+Price
83	Dry grocery	Spices, Seasoning, Extracts	Seasoning-Dry; spice = Garlic powder/granules	Spice	100%		No		
84	Dry grocery	Spices, Seasoning, Extracts	Seasoning-Dry; spice = Meat seasoning	Spice	100%		Yes	GH+H	1+PC+Price
85	Dry grocery	Spices, Seasoning, Extracts	Vegetables - Onions - Instant	Chopped; minced	75%	Dry onions	Yes	GH	1+PC+Price
86	Dry grocery	Vegetables - Canned	Mushrooms - Shelf Stable	Cut type; salt; type of mushroom	40%		No		
87	Dry grocery	Vegetables - Canned	Vegetables-Mixed-Canned	Brand tier; low salt; preparation style; vegetable type	28%	Beans city average x vegetable type == "bean"; tomatos x vegetable type contains "tomato"	Yes	GH	1+PC+Price
88	Dry grocery	Vegetables - Canned	Vegetables-Peas & Carrots-Canned; Vegetables-Peas-Canned; Vegetables-Peas-Remaining-Canned	Has carrots; has meat; salt; type of peas	49%	Aluminum; carrots x has carrots; green peas x type of peas != "blackeye"; salt pepper spices x salt	Yes	GH	1+PC+Price
89	Dry grocery	Vegetables And Grains - Dried	Rice - Instant	Instant	79%	Fertilizer; rough rice	Yes	GH	1+PC+Price
90	Frozen foods	Baked Goods-Frozen	Bakery-Bagels-Frozen	Bagel size; has no salt; is low fat; is sliced; is wheat; primary flavor; quality tier; secondary flavor	32%	Sugar x primary flavor == "sweet"; wheat; wheat flour	Yes	GH	1+PC+Price
91	Frozen foods	Breakfast Foods-Frozen	Frozen/Refrigerated Breakfasts	Breakfast type; carb; has cheese; healthy; is veg; protein	26%	Eggs x protein == "egg"; poultry processing x protein in ["chicken", "turkey"]; slaughter cattle x protein == "beef"; slaughter hogs x protein in ["ham/bacon", "sausage"]; wheat x carb != "other"	Yes	GH+CS	1+PC+Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
92	Frozen foods	Pizza/Snacks/Hors D'oeuvres-Frzn	Pizza-Frozen	Crust; healthy; microwave; quality; topping	26%	Cheese; slaughter cattle x topping in ["pepperoni", "other meats"]; slaughter hogs x topping in ["pepperoni", "other meats"]; tomatos; wheat; wheat flour	Yes	GH	1+PC+Price
93	Frozen foods	Unprep Meat/Poultry/Seafood-Frzn	Frozen Poultry	Boneless; chicken; cut; meat	34%	Poultry processing; processed foods and feeds; slaughter poultry	Yes	GH	1+PC+Price
94	General merchandise	Kitchen Gadgets	Beverage Storage Container	Bottle type; quality; size description; volume oz	34%	Plastic; stainless steel	Yes	GH	1+PC+Price
95	General merchandise	Stationery, School Supplies	Dry Erase Bulletin Board And Accesory	Board type; has calendar; has cork; is magnetic	47%		Yes	GH+H	1+PC+Price
96	General merchandise	Stationery, School Supplies	Personal Planners Binders And Folders	Product type	35%	Plastic; pulp paper	Yes	GH	1+Price
97	Health & beauty care	Cosmetics	Cosmetic Kits	Kit type	42%		Yes	GH	1+PC+Price
98	Health & beauty care	Cosmetics	Cosmetics - Concealers	Product type	50%		No		
99	Health & beauty care	Cosmetics	Cosmetics-Blushers	Blush type	61%	Fatty acids; starch vegetable fats oils	No		
100	Health & beauty care	Cosmetics	Cosmetics-Eye Shadows	Quality	58%		No		
101	Health & beauty care	Cosmetics	Cosmetics-Eyebrow & Eye Liner	Product type	71%		Yes	GH	1+PC+Price
102	Health & beauty care	Cosmetics	Cosmetics-Face Powder	Skin type	70%		No		
103	Health & beauty care	Cosmetics	Cosmetics-Foundation-Cream And Powder; Cosmetics-Foundation-Liquid	Foundation type	55%		No		
104	Health & beauty care	Cosmetics	Cosmetics-Lipsticks	Product purpose	45%	Fatty acids; starch vegetable fats oils	Yes	GH	Price
105	Health & beauty care	Cosmetics	Cosmetics-Mascara	Features	38%		No		
106	Health & beauty care	Cosmetics	Talcum & Dusting Powder	Scent	66%	Corn starch; talc	Yes	GH+H	Price
107	Health & beauty care	Fragrances - Women	Cologne & Perfume-Women's	Common consumer name; value	46%	Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"	No		
108	Health & beauty care	Grooming Aids	Cosmetic And Nail Grooming Accessory	Primary type	53%		Yes	GH	Price
109	Health & beauty care	Hair Care	Hair Preparations - Other Than Men's	Form; use	29%	Basic organic compounds; ethanol	No		
110	Health & beauty care	Hair Care	Hair Spray - Women's	Aerosol; scented	73%	Basic organic compounds; ethanol	No		
111	Health & beauty care	Hair Care	Shampoo-Aerosol/ Liquid/ Lotion/ Powder; Shampoo-Combinations	Dandruff; liquid; regular or other; two in one	43%	Fatty acids; surfactants	No		
112	Health & beauty care	Medications/Remedies/Health Aids	Foot Preparations-Athlete's Foot	Cream; deodorant; extra strength; liquid; liquid spray; ointment; powder; powder spray	41%	Basic organic compounds; basic organic compounds x extra strength == 1	Yes	GH	Price

Merger	NielsenIQ department	NielsenIQ product group	Product market definition	Characteristics	% variance explained by two PCs	Cost shifters	In estimation sample	Instruments used	Demographic interactions
113	Health & beauty care	Men's Toiletries	Cologne/Lotion-Men's	Common consumer name	53%	Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"	No		
114	Health & beauty care	Men's Toiletries	Cologne/Lotion-Men's	Common consumer name; form; value	41%	Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"	Yes	GH+H	1+Price
115	Health & beauty care	Skin Care Preparations	Hand Cream	High price tier; scent; target condition	49%	Fatty acids; starch vegetable fats oils	No		
116	Health & beauty care	Skin Care Preparations	Skin Cream-All Purpose	High price tier; scent; target condition; vitamin e	44%	Fatty acids; starch vegetable fats oils; vitamins x vitamin e == 1	No		
117	Non-food grocery	Detergents	Detergents - Heavy Duty - Liquid; Detergents - Light Duty; Detergents-Packaged	Scent none	70%	Surfactants	No		
118	Non-food grocery	Laundry Supplies	Detergent Boosters	Liquid; powder	67%	Surfactants	Yes	GH	1+PC+Price
119	Non-food grocery	Tobacco & Accessories	Cigarettes	Low tar; menthol	74%	Pulp paper; tobacco	Yes	GH+CS	1+PC+Price
120	Packaged meat	Packaged Meats-Deli	Bratwurst & Knockwurst; Frankfurters-Refrigerated; Sausage-Dinner	Form; frozen; hot; meat; sausage type; skinless	14%	Beef and veal x meat contains "(beef veal)"; poultry processing x meat contains "(chicken turkey)"; processed meat; salt pepper spices; slaughter cattle x meat contains "beef"; slaughter hogs x meat contains "pork"; slaughter poultry x meat contains "(chicken turkey)"	Yes	GH+CS	1+PC+Price
121	Packaged meat	Packaged Meats-Deli	Bratwurst & Knockwurst; Frankfurters-Refrigerated; Sausage-Dinner	Form; frozen; hot; meat; sausage type; skinless	16%	Beef and veal x meat contains "(beef veal)"; poultry processing x meat contains "(chicken turkey)"; processed meat; slaughter cattle x meat contains "beef"; slaughter hogs x meat contains "pork"; slaughter poultry x meat contains "(chicken turkey)"	Yes	GH	1+PC+Price
122	Packaged meat	Packaged Meats-Deli	Lunchmeat-Sliced-Refrigerated	Flavor; form; low fat; low salt; meat type	25%	Beef and veal x meat type contains "(beef veal)"; poultry processing x meat type in ["turkey", "chicken"]; processed meat; salt pepper spices; slaughter cattle x meat type in ["beef", "salami pepperoni"]; slaughter hogs x meat type in ["ham", "salami pepperoni"]; slaughter poultry x meat type in ["turkey", "chicken"]	Yes	GH	Price
123	Packaged meat	Packaged Meats-Deli	Sausage-Breakfast	Flavor; form; low fat; meat; precooked	21%	Beef and veal x meat contains "(beef veal)"; poultry processing x meat contains "(chicken turkey)"; processed meat; salt pepper spices; slaughter cattle x meat contains "beef"; slaughter hogs x meat contains "pork"; slaughter poultry x meat contains "(chicken turkey)"	Yes	GH	1+PC+Price

Table B.1: All mergers in our sample. Characteristics explained by principal components always include an indicator for whether the product is a store brand, and the package size. The possible values for "Instruments used" are "GH" (only differentiation instruments), "GH+CS" (differentiation and cost shifter instruments), and "GH+H" (differentiation and Hausman instruments). The possible values for "Demographic interactions" are "Price" (high income indicator interacted with only price), "1+Price" (interacted with price and the constant), and "1+PC+Price" (interacted with price, the constant, and principal components of characteristics).